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DEPARTMENT OF COMPUTER SCIENCE



M-Health Application for Tuberculosis Screening and Monitoring in rural communities of Eswatini

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A research project submitted to the University of Eswatini in partial fulfilment of the requirements for a Bachelor's Degree in Computer Science Education.

DECLARATION

We, Maseko Mayenziwe (202100599) and Mndzebele Sinethemba (202102919), do hereby declare that this dissertation, submitted to fulfil the obligation for the Bachelor of Science in Computer Science Education degree, represents our original work and that due attribution has been provided for all sources cited in supporting our research.

This dissertation was prepared in accordance with the University of Eswatini's standards and regulations with regard to originality and academic quality in degree programs, the research was conducted in accordance with the ethical rules and principles of the Institute.

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ABSTRACT

Tuberculosis (TB) remains a significant public health challenge in Eswatini, particularly in rural communities where limited access to healthcare, diagnostic delays, and stigma hinder early detection and treatment. This study aimed to design, develop, and evaluate a mobile health (mHealth) application to enhance TB screening and monitoring in these underserved areas. Employing a mixed-methods approach, the research combined quantitative data from structured surveys and system usage logs with qualitative insights gathered through user feedback and usability testing. The mHealth application was developed using Flutter for the frontend and Firebase for the backend, and it features a digital TB symptom screening tool, referral system using geolocation, multilingual educational content, and offline functionality. A provider dashboard was also implemented to support healthcare workers in managing referrals and monitoring patient data. Evaluation results indicated high usability (mean SUS score of 84.5), strong offline engagement, and favourable user acceptance. Qualitative feedback emphasized the importance of privacy, security, and educational support. The findings demonstrate that mHealth technology can be effectively used to overcome structural and informational barriers to TB control in Eswatini. The application shows strong potential for integration into national TB programs and for expansion to address other public health challenges in low-resource settings.

ACRONYMS

Acronym	Meaning
API	Application Programming Interface
CDC	Centers for Disease Control and Prevention
HER	Electronic Health Records
HIV	Human Immunodeficiency Virus
MDR-TB	Multidrug-Resistant Tuberculosis
mHealth	Mobile Health
N/A	Not Applicable
NGO	Non-Governmental Organization
NTCP	National Tuberculosis Control Programme
SDLC	Software Development Life Cycle
SUS	System Usability Scale
TB	Tuberculosis
UHC	Universal Health Coverage
UI	User Interface
WHO	World Health Organization

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1. INTRODUCTION

1.1. Introduction

Tuberculosis (TB) remains one of the world's most lethal infectious diseases, claiming approximately 1.3 million lives every year and affecting millions more, particularly in low-income countries (World Health Organization, 2023). In Eswatini specifically, the TB epidemic presents a formidable public health challenge. TB had an incidence rate of 363 cases per 100,000 population as of 2022, positioning the country among the 30 high TB/HIV burden nations globally (World Health Organization, 2023). This situation is made more complicated by the concurrent HIV epidemic, as seen by the high HIV-TB co-infection rates. They stand at approximately 68%, significantly increasing morbidity and mortality rates (Reid et al., 2019).



Figure 1: Estimated TB incidence rates, 2022

The burden of TB is disproportionately felt in rural communities of Eswatini, where multiple barriers contribute significantly to ineffective disease management. These communities face numerous challenges including limited access to healthcare facilities, inadequate transportation infrastructure, shortage of healthcare personnel, and insufficient diagnostic resources (Haumba et al., 2015). As a result, rural residents often experience delayed diagnosis, with studies indicating diagnostic delays averaging 4-8 weeks in similar sub-Saharan African settings (Getnet et al., 2017). These delays not only result in worse

individual health outcomes. They also facilitate continued community transmission, perpetuating the epidemic cycle.

Mobile health (mHealth) technologies have emerged as promising tools for addressing healthcare delivery challenges in resource-limited settings. There exists significant potential to leverage this technology to enhance TB control efforts in Eswatini. Research has demonstrated that mHealth interventions can effectively improve TB screening uptake, treatment adherence, and health outcomes in similar contexts (Keutzer et al., 2020a). By extending the reach of healthcare services beyond traditional facility-based approaches, mHealth solutions can help overcome geographic, financial, and system-level barriers that currently affect TB control efforts.

This research is focused on the development and implementation of a comprehensive mHealth application specifically designed to address TB screening and monitoring challenges in rural Eswatini. The proposed application incorporates several key features, namely a digital symptom screening tool that connects users with healthcare professionals, geolocation services for health facility referrals, user accounts for personalized interactions, educational content to improve TB knowledge, multilingual support to address communication barriers, and offline functionality to overcome connectivity limitations in rural areas. Additionally, the application includes a healthcare provider dashboard for patient management and demographic data analysis to support evidence-based program improvements.

Our study begins by examining the current TB situation in Eswatini, with particular attention to rural communities, followed by a detailed discussion of existing gaps in TB screening and monitoring services. We present the methodology for developing and implementing the proposed mHealth application, including user-centred design approaches and technical specifications. The study then proceeds to outline the implementation strategy, evaluation framework, and anticipated outcomes. Finally, we discuss the potential implications of this research for TB control efforts in Eswatini and similar settings, as well as considerations for sustainability and scalability.

1.2. Background of study

Tuberculosis (TB) remains a significant public health challenge in Eswatini, a country grappling with one of the highest TB incidence rates in the world. According to Eswatini Ministry of Health (2024b), Eswatini has an estimated TB incidence rate of 733 cases per 100000 population, with a prevalence of approximately 907 cases per 100000. This alarming situation is exacerbated by the high co-infection rates with HIV, where over 80% of TB patients are also living with HIV, making it one of the 30 high burden countries globally (World Health Organization, 2023). The intersection of these two epidemics poses a unique challenge to the healthcare system, necessitating innovative approaches to enhance TB detection and treatment.

The red square indicates that a country is in a list.

COUNTRY	TB	TB/HIV	MDR/RR-TB
Angola	■		■
Azerbaijan			■
Bangladesh	■		■
Belarus			■
Botswana		■	
Brazil	■		■
Cameroon		■	
Central African Republic	■	■	
China	■	■	■
Congo	■	■	
Democratic People's Republic of Korea	■		■
Democratic Republic of the Congo	■	■	■
Eswatini		■	
Ethiopia	■	■	
Gabon	■	■	
Guinea		■	
Guinea-Bissau		■	
India	■	■	■
Indonesia	■	■	■
Kazakhstan			■
Kenya	■	■	
Kyrgyzstan			■
Lesotho	■	■	
Liberia	■	■	
Malawi		■	
Mongolia	■		■
Mozambique	■	■	■
Myanmar	■	■	■
Namibia	■	■	
Nepal			■
Nigeria	■	■	■
Pakistan	■		■
Papua New Guinea	■		■
Peru			■
Philippines	■	■	■
Republic of Moldova			■
Russian Federation		■	■
Sierra Leone	■		
Somalia			■
South Africa	■	■	■
Tajikistan			■
Thailand	■	■	
Uganda	■	■	
Ukraine			■
United Republic of Tanzania	■	■	
Uzbekistan			■
Viet Nam	■		■
Zambia	■	■	■
Zimbabwe		■	■

Figure 2: Countries in the three global lists of high-burden countries for TB, HIV-associated TN and <DR/RR-TB to be used by WHO in the period 2021-2025

The National Tuberculosis Control Programme (NTCP) in Eswatini has implemented various strategies to combat TB, including integrating TB screening into maternal and child health services and targeting high-risk populations such as prisoners and pregnant women (Stop TB Partnership, 2014). For instance, NTCP deployed TB Screening Officers at antenatal clinics to ensure systematic screening of all pregnant women accessing services. This initiative aimed to detect an additional 1181 TB from a projected 83059 suspects screened annually (Stop TB Partnership, 2014). Moreover, the introduction of rapid diagnostic tools like GeneXpert has significantly improved the accuracy and speed of TB diagnosis in these vulnerable populations (Stop TB Partnership, 2014).

Despite the efforts, significant gaps remain in case detection and treatment adherence. The current case detection rate stands at only 60%, indicating that many individuals remain undiagnosed or untreated (Eswatini Ministry of Health, 2024b). TB treatment coverage stands at only 68%, falling short of WHO's target of 90% by 2030 (Eswatini Ministry of Health, 2024a). The challenges are further compounded by limited healthcare infrastructure in rural areas, where access to diagnostic services is often inadequate. This lack of access contributes to delayed diagnoses and increased transmission rates within communities.

The economic implications of TB are profound. Patients often face substantial out-of-pocket expenses for healthcare services, which can lead to financial hardship and push families deeper into poverty. Additionally, untreated TB can result in lost productivity and increased healthcare costs for both families and the national health system.

In light of these challenges, the NSP outlines strategic objectives aimed at achieving a TB-free Eswatini by 2035. Key goals include reducing TB incidence by 40% relative to 2022 levels and achieving a treatment success rate of 90% for all forms of TB by 2028 (Eswatini Ministry of Health, 2024a). To meet these ambitious targets, the NSP emphasizes leveraging new technologies and adopting a patient-centered approach to care. There is an urgent need for innovative solutions that leverage technology to enhance TB screening and monitoring in rural areas. mHealth applications present a promising avenue for addressing these issues by providing accessible and efficient means of screening, patient management and adherence tracking.

The current context underscores the importance of this research. As communities recover from the disruptions caused by the COVID-19 pandemic, there is a critical need to strengthen

public health efforts against TB. mHealth solutions can play a pivotal role in ensuring health outcomes improve for those affected by this disease.

In summary, addressing the dual challenges of TB and HIV in Eswatini requires a multifaceted approach that includes innovative technology solution. By focusing on enhancing screening and treatment adherence through these platforms, this research aims to contribute significantly to improving public health outcomes in rural communities

1.3. Problem Statement

TB is one of the world's most deadly infectious diseases, causing an estimated 1.25 million deaths and 10.8 million new infections in 2023 (World Health Organization, 2023). Despite being preventable and treatable, TB continues to claim approximately 3,500 lives daily, largely due to delayed diagnosis, poor treatment adherence, and limited access to healthcare. These challenges are particularly severe in rural and poorly supported populations, where one undiagnosed case can infect 10 to 15 others annually, increasing the cycle of transmission.

Eswatini is among the countries most affected by TB, with one of the highest TB incidence rates worldwide. Although the prevalence of TB cases has declined from 1,350 per 100,000 people in 2012 to 363 per 100,000 in 2019, TB-related mortality remains high. According to Kerschberger et al. (2024), more than 10% of TB patients die during treatment, often due to late diagnosis, poor adherence to treatment, and limited access to health services in rural areas. Stigma around TB further worsens the situation, as many people avoid sharing their diagnosis with family members, which delays treatment and lowers the chances of recovery as stated by Dormechele et al. (2024). These issues show the need for targeted interventions in Eswatini's rural areas to address the high death rates from TB.

Digital tools, such as mobile health (m-health) applications, have demonstrated potential in enhancing TB care by supporting early diagnosis, encouraging treatment adherence, and reducing stigma. M-health tools also include features to identify individuals at risk of TB and offer preventive care. Despite this potential, m-health solutions are not widely used in Eswatini for TB management, leaving a gap in using technology to improve care as stated by (Centers for Disease Control and Prevention, 2024; World Health Organization, 2023).

This study aims to develop and evaluate a m-health application designed specifically for rural communities in Eswatini. The proposed app will focus on TB screening, monitoring, and

reducing stigma, while also improving and early diagnosis. By addressing these challenges, the project aims to lower TB-related deaths and support the goals of the WHO's End TB Strategy to achieve better health outcomes in limited access populations.

1.4. Aims and objectives

The aim of this study is to design, develop and evaluate an M-Health application aimed at improving TB screening and monitoring in rural areas.

Objectives of the study are to:

1. Review current TB screening and monitoring systems in rural communities of Eswatini
2. Develop an M-Health application for TB screening and monitoring
3. Implement and evaluate proposed M-Health application

1.5. Research questions

1. What are the current TB screening and monitoring systems in rural communities of Eswatini?
2. How can an M-Health application be developed for TB screening and monitoring in rural communities of Eswatini?
3. How can the proposed M-Health applications for TB screening and monitoring be implemented and evaluated in rural communities of Eswatini?

1.6. Justification of the study

The proposed research on developing and implementing an M-Health application for TB screening and monitoring in rural areas of Eswatini addresses significant public health challenges which are worsened by the intersection of the TB and HIV epidemics.

As earlier mentioned, TB is grappling with one of the highest TB incidence rates globally. Despite ongoing efforts by NTCP, including initiatives to integrate TB screening into maternal and child health services, the case detection rate remains insufficient. The existing healthcare infrastructure in rural areas often lacks the necessary resource and trained personnel to conduct systematic screenings effectively. This situation is further complicated by high rates of co-infection with HIV, which significantly increases TB morbidity.

Importance of solving the problem

The implications of addressing these gaps through an mHealth application are significant. Early detection and consistent treatment monitoring can significantly reduce TB transmission

rates and improve individual health outcomes. Research indicates that each undiagnosed active TB case can infect 10-15 people annually, making effective screening crucial for disease control (World Health Organization, 2023). Also, TB imposes substantial economic burdens on affected households and the healthcare system. Studies from similar settings in sub-Saharan Africa estimate that TB-affected households lose approximately 58-85% of their annual income due to the disease (Tanimura et al., 2014). By improving early detection, the economic burden can be substantially reduced.

The dashboard functionality for healthcare staff, along with demographic analysis capabilities, addresses system-level inefficiencies. Health systems research indicates that digital health interventions can improve healthcare resource allocation by 30-45% through better data management and targeted interventions (Agarwal et al., 2017).

Research gaps

The proposed mHealth application addresses several critical gaps in the current TB management approach in rural Eswatini. Firstly, the digital screening tool feature will facilitate systematic symptom screening without requiring immediate physical access to healthcare facilities. By enabling users to input health information that is then evaluated by qualified health staff, the application bridges the gap between rural communities and healthcare expertise. This approach aligns with WHO's recommendation for innovative case-finding strategies in high-burden settings.

The geolocation feature for health facility referral directly addresses the geographic barriers faced by rural communities. Research indicates that distance to health facilities is a significant predictor of delayed TB diagnosis, with patients living more than 10 km from health facilities being 57% less likely to seek timely care (Gebeyehu et al., 2014). The ability to identify and navigate to the nearest appropriate healthcare facility can substantially reduce these delays.

The educational content component addresses the critical need for TB awareness and health literacy in rural communities. Studies have shown that inadequate knowledge about TB symptoms and transmission contributes to diagnostic delays averaging 4-8 weeks in sub-Saharan Africa (Getnet et al., 2017).

Multilingual support is essential in Eswatini, where both siSwati and English are official languages, and various dialects are spoken in rural areas. Language barriers have been

identified as significant obstacles to effective healthcare delivery, with studies showing that patients who receive health information in their native language demonstrate 35% higher adherence rates (Nglazi et al., 2013).

The offline functionality feature directly addresses the reality that a substantial number of people in rural areas in Eswatini lack reliable internet connectivity. This feature ensures continued access to vital TB information and screening tools regardless of connectivity status.

Why the study is unique

The study has a comprehensive feature set, combining features like digital screening, geolocation services, educational content, multilingual support, offline functionality, and user accounts. This represents a holistic approach to addressing multiple barriers simultaneously.

In addition, unlike generic mHealth solutions, this application is specifically tailored to the unique challenges and needs of rural Eswatini, considering local language requirements, connectivity limitations, and health system structures.

The application is also designed to complement and enhance existing NTCP efforts rather than replace them, facilitating better integration with the current healthcare infrastructure.

Beneficiaries of this study

The primary beneficiaries of this study include:

- **TB Patients and At-Risk Individuals:** Rural community members will benefit from improved access to screening, education, and referral services.
- **Healthcare Providers:** The staff dashboard and referral tracking features will enhance healthcare workers' ability to manage patients effectively and allocate resources efficiently.
- **Health System:** The data generated through the application will provide valuable insights for program planning and resource allocation, potentially improving the efficiency and effectiveness of TB control efforts.

In conclusion, the proposed mHealth application represents a timely, comprehensive, and contextually relevant intervention to address significant gaps in TB screening and monitoring in rural Eswatini. Its timing is particularly crucial as health systems recover from COVID-19 disruptions. Recent data shows that TB deaths increased globally for the first time in over a

decade due to pandemic-related service interruptions (World Health Organization, 2023). Innovative digital solutions have become increasingly important in the post-pandemic context, with WHO identifying digital health as a key strategy for rebuilding and strengthening TB services (World Health Organization, 2023). By leveraging mobile technology to overcome geographical, knowledge, and system-level barriers, this research has the potential to significantly contribute to improved TB control and better health outcomes for affected populations.

1.7. Chapter Summary

This chapter has outlined the critical public health challenge posed by tuberculosis (TB) in Eswatini, particularly in rural communities where the disease burden is compounded by high rates of HIV co-infection and limited healthcare resources. Despite national efforts to improve TB detection and treatment, significant gaps remain in early diagnosis, treatment adherence, and access to care, especially in underserved areas. The chapter highlighted the potential of mobile health (mHealth) technologies to address these barriers by providing innovative, accessible solutions for TB screening, monitoring, and education. The proposed research aims to design, develop, and evaluate a tailored mHealth application that integrates digital screening tools, geolocation services, educational resources, multilingual support, and offline functionality to enhance TB control efforts in rural Eswatini. By bridging gaps in healthcare delivery and supporting both patients and providers, this study seeks to contribute to national and global TB elimination goals, offering a scalable model for similar high-burden, resource-limited settings.

2. LITERATURE REVIEW

2.1. Introduction

TB remains one of the foremost public health challenges in many developing countries, and Eswatini is one of them (World Health Organization, 2023). Despite global efforts to implement effective screening and treatment, many regions continue to rely on traditional, paper-based methods that are both time-consuming and prone to errors. With advancements in mobile technology and increasing smartphone penetration even in rural areas, there is a compelling case to review and consider m-health applications as a feasible alternative in improving TB screening and monitoring. This review critically analyses existing literature on m-health solutions, focusing on their evolution, benefits, challenges, and specific applicability within the rural communities of Eswatini.

This literature review demonstrates that m-health applications can not only augment the current paper-based system in Eswatini but also lead to more efficient TB screening, timely monitoring, and ultimately enhanced health outcomes. Even in cases where infrastructure and cultural challenges seem prohibitive, evidence from recent studies suggests there is great potential for scalable mobile health solutions.

2.2. Evolution of M-Health Applications in Healthcare

The concept of mobile health (m-health) has evolved significantly over the past decade, moving from rudimentary SMS-based reminders and appointment scheduling to comprehensive platforms that integrate diagnostic support, data management, and real-time monitoring. According to World Health Organization (2021), mobile health (mHealth), defined as the use of mobile and wireless technologies to support the achievement of health objectives, has rapidly expanded over the past decade, transforming how health care is delivered and accessed globally, particularly in low - and middle-income countries. The literature demonstrates that m-health applications are not only gaining popularity in high-income countries but are increasingly recognized for their potential in low-resource environments.

The more modern concept of mHealth emerged in the early 2000s, initially focusing on basic health services such as SMS reminders for medication adherence. However, it was the introduction of smartphones, particularly the iPhone in 2007, that marked a significant

turning point in mHealth development. Smartphones enabled the creation of complex apps capable of performing a wide range of health-related functions, from monitoring vital signs to facilitating remote consultations.

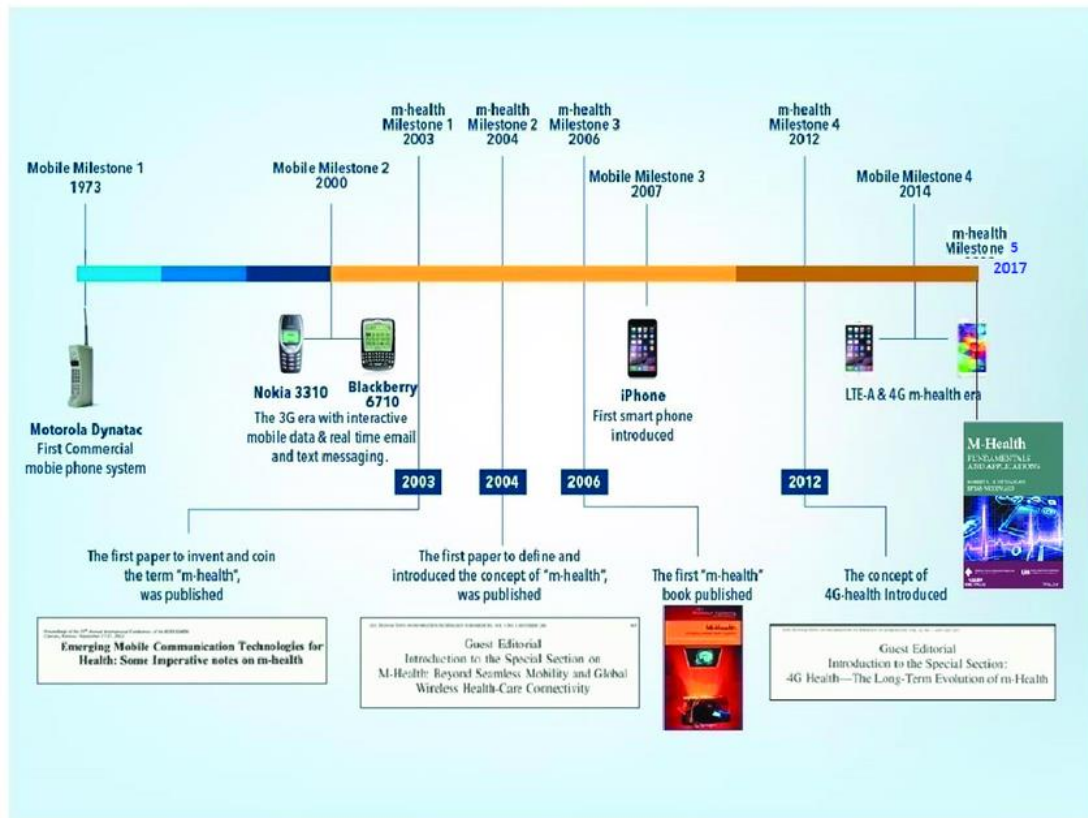


Figure 3: The historical perspective of the evolution of mobile health

In the context of TB management, m-health applications have evolved to include modules that support symptom screening questionnaires, digital chest radiograph interpretation, and automated referral systems. Recent studies indicate that these systems can facilitate early diagnosis, reduce loss-to-follow-up, and optimize the allocation of scarce healthcare resources (Shah et al., 2022). Moreover, the evolution of computer technology and, in particular, computational power, has led to the development of specific computational scientific disciplines such as bioinformatics and systems biology as stated by Cuff (2023). Notably, various pilots in sub-Saharan Africa, including clinical trials and field evaluations, have underscored the feasibility and adaptability of m-health tools in rural settings.

In Eswatini, while the traditional TB screening system remains largely paper-based, recent small-scale projects illustrate an emerging trend of digital innovation as patients are now recorded in a computer system. The transitional phase of integrating mobile technology amidst predominantly manual record-keeping poses both unique opportunities and challenges. Although smartphone penetration has historically been low in rural communities, recent governmental and public-private partnership initiatives have shown a steady increase in availability and usage (Doyle et al., 2021). Although there is no clear evidence of availability of mobile phone in rural areas of eSwatini. This evolution is critical as it signals readiness for transitioning from manual to digitized health interventions.

**Ministry of Health
TB Screening Tool**

Name of Facility:		Unit:	
Patient Data		Adults	
Date: / /		Symptom 1	Symptom 2
High Risk Population ☐		Current Cough? (a) ☐	Fever (b) ☐
Entry Point ☐		Symptom 3	
Age:		Night Sweats (c) ☐	
Patient ID:		Symptom 4	
Sex: M ☐ F ☐		Unexplained weight loss? (d) ☐	
Weight (in Kgs):		Children < 5	
Height (m):		Current Cough? (a) ☐	Fever (b) ☐
ACTION TAKEN →		Poor weight gain/failure to thrive (c) ☐	
		History of contact with a TB case? (d) ☐	
→ If Yes to "ONE" of the 4 symptoms consider that as a presumptive case → For Children <5 years, History of Contact with TB case alone should be considered presumptive TB case. HIGH RISK POPULATION STATUS: 1. Miners 2. Ex-miners 3. PLHV 4. Prisoners 5. Contact of person with TB 6. Health Care Worker 7. Seasonal Workers 8. Migrants 9. Factory Worker 10. People with NCDs 11. Children Under 5 12. Family member of miner or ex-miner 13. Other			
SCREEN RESULT		POSITIVE (Presumptive TB)	
Sputum collected for Lab Investigation Yes ☐		Referred to Doctor for Clinical Assessment Yes ☐	
No ☐		No ☐	
NEGATIVE		SCREEN RESULT	

Figure 4: Ministry of Health TB paper-based screening tool

Furthermore, the evolution of m-health applications in healthcare has been underpinned by global initiatives aimed at achieving universal health coverage (UHC) and the Sustainable Development Goals (SDGs), particularly in relation to TB control. Enhanced connectivity, cost-effective solutions, and improved data accessibility underline the emerging consensus in

the literature that m-health can address many of the systemic inefficiencies in traditional healthcare systems (Njoku, 2021). Thus, the progression of m-health from a supplementary tool to a cornerstone in healthcare delivery represents a paradigm shift that is highly relevant for the TB screening protocols in Eswatini.

2.3. Benefits of M-Health Applications in TB Screening and Monitoring

Integrating m-health applications into TB screening and monitoring introduces numerous advantages compared to the traditional paper-based methods. One of the main benefits is enhanced data accuracy and real-time data transmission. Automated data capture minimizes human error, ensures consistency, and facilitates prompt decision-making in clinical settings (Maleki Varnosfaderani & Forouzanfar, 2024). Mobile applications can integrate diagnostic algorithms that systematize symptom evaluation, thereby providing frontline healthcare workers with immediate feedback on risk levels.

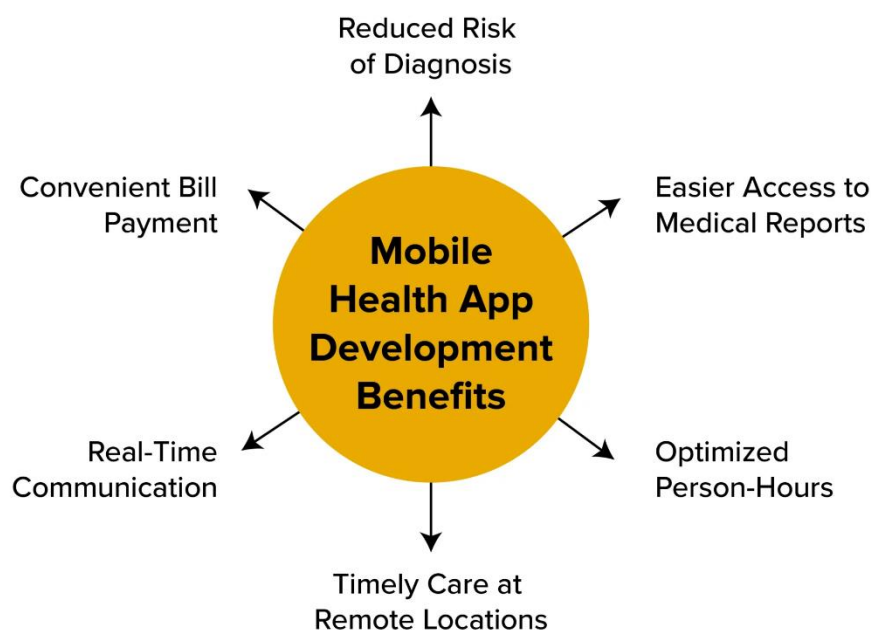


Figure 5: Benefits of mobile health apps for healthcare professionals and patients

Another key benefit lies in the improved accessibility and convenience for both patients and healthcare providers. According to Okolo, Babawarun, & Olorunsogo (2024), M-health solutions allow for remote monitoring of patients, enabling continuous follow-up even in geographically isolated communities. This is particularly critical in rural Eswatini, where

physical distance and limited transport options hamper regular clinic visits. The use of m-health platforms can transform patient engagement by providing reminders, educational content, and interactive interfaces that empower individuals to manage their health proactively (Okolo, Babawarun, Arowoogun, et al., 2024).

Cost-effectiveness is a further advantage of m-health interventions. When compared to extensive infrastructure investments required for traditional healthcare delivery, m-health platforms can reduce costs related to travel, paper-based record keeping, and repetitive manual processes. The consolidation of data into centralized digital records also allows for better resource planning and equitable allocation of limited diagnostic and treatment resources. According to recent literature by Abernethy et al. (2022), resource optimization through digitization not only streamlines operations but also results in long-term savings for national health systems.

Moreover, the capability for rapid and systematic analysis of large datasets is another pivotal benefit. M-health applications can aggregate data across different regions and time periods, providing a comprehensive overview of TB trends in real time. This facilitates a proactive public health response through targeted interventions, early outbreak detection, and efficient contact tracing protocols (Yousif et al., 2024). With the implementation of advanced data analytics, healthcare systems gain a deeper insight into disease patterns, enabling better decision-making at policy and operational levels.

The literature also highlights the role of m-health in strengthening community health initiatives. Involving local health workers through training and providing them with digital tools has positive implications for community engagement as stated by Fitzpatrick (2023). Gonzalez et al. (2021) emphasized that by embedding culturally sensitive content and local language support into applications, m-health platforms can enhance the acceptability and effectiveness of TB interventions in developing countries like eSwatini diverse cultural landscape. This community-level approach is crucial for fostering public trust and ensuring high rates of adoption among rural populations.

2.4. Challenges of M-Health Applications in TB Screening and Monitoring

Despite its numerous benefits, the application of m-health in TB screening and monitoring is accompanied by significant challenges. One primary concern is the digital divide that exists in many rural communities, particularly the gap in smartphone penetration and reliable mobile network coverage as stated by Jongebloed et al. (2024). In Eswatini, although there has been progress in mobile connectivity, rural areas often continue to experience discrepancies in network reliability when compared to urban settings. These limitations can hinder the real-time functioning of m-health applications and may delay critical interventions. Keutzer et al. (2020b) conducted a descriptive review on M Health applications aimed at improving TB treatment. The study analyzed various mobile apps designed to support TB patients and healthcare providers by enhancing treatment adherence, monitoring progress, and providing education.

Meyer et al. (2020) conducted a case study on implementing mHealth interventions in Uganda, a resource-constrained setting. The study explored the challenges and opportunities associated with deploying mobile health solutions for tuberculosis (TB) care. Key barriers included limited smartphone access, poor internet connectivity, and concerns over data privacy.

The integration of m-health technologies also requires robust infrastructure and technical support, which are not uniformly available as stated by Giebel et al. (2023). The deployment and sustainability of digital platforms depend on a continuous supply of electricity, data services, and technical expertise which are often lacking in remote settings. Additionally, data security and patient privacy are paramount concerns. In many instances, the rapid deployment of m-health solutions means that cybersecurity measures lag behind, exposing sensitive health information to potential breaches. Healthcare institutions must therefore invest in stringent cybersecurity protocols to safeguard patient data, which inevitably increases operational costs.

Socio-cultural factors present another significant challenge. In rural Eswatini, where traditional practices and beliefs are deeply rooted, acceptance of digital health technologies is not uniform. Resistance may arise due to a distrust of modern technology, fear of change, or concerns about the relevance of digital interventions in a culturally specific context. Moreover, literacy levels and language barriers can affect the effective usage of these

applications, necessitating the development of user-friendly interfaces that cater to different educational.

There are also logistical challenges linked to the transition away from paper-based systems. Health workers accustomed to conventional practices may require intensive training and supportive supervision to adapt to new digital routines. This shift not only demands financial investment but also comprehensive change management strategies to ensure that healthcare professionals are equipped to use m-health applications confidently and effectively. Resistance to change is often amplified by the perception that digital systems may complicate routines rather than simplify them, particularly in institutions where resources for additional training may be scarce.

Lastly, the sustainability of m-health solutions remains a critical challenge. Pilot projects done by Chaiyachati et al. (2013) often show promising results, but scaling these interventions to regional or national levels introduces a host of issues, including maintenance, funding continuity, and technological upgrades. It is essential that the cost implications, both upfront and ongoing, are carefully analyzed and planned for so that m-health interventions remain viable in the long term. Without adequate financial and institutional support, the initial surge in digital solution adoption might prove ephemeral as older systems fail to keep pace with technological advances or operational demands.

2.5. Gaps in Existing Research

Despite the growing body of literature supporting the feasibility of m-health applications in improving public health outcomes, significant research gaps remain, particularly in the context of rural Eswatini. One major area of under-exploration is the comprehensive assessment of cultural and social factors that influence m-health adoption. While studies have highlighted general barriers, there is limited in-depth qualitative research examining how local attitudes, beliefs, and traditions specifically affect patient and healthcare provider engagement with m-health platforms. There is a lack of comprehensive exploration of socio-economic barriers, particularly the impact of gender dynamics on m-health adoption in rural sub-Saharan Africa (Kumah, 2025). The gap is especially relevant for culturally diverse settings like Eswatini, where social roles may influence technology acceptance. Such insights are critical to designing culturally sensitive applications that resonate with local communities and address inherent biases against digital health solutions.

Another notable gap pertains to the integration of m-health with existing healthcare infrastructures. While the interoperability of digital systems is frequently mentioned, few studies have empirically examined the process of merging new m-health platforms with pre-existing paper-based systems. Research is needed to understand the challenges of digital data migration, synchronization issues, and the potential for redundancy or data loss when transitioning from analog to digital records.

There is also a shortage of longitudinal studies documenting the long-term outcomes of m-health interventions in TB screening and monitoring. According to Sekandi et al. (2025), an insufficient understanding of user engagement metrics and their influence on long-term adherence to m-health TB interventions is the cause. Most existing research focuses on short-term outcomes, leaving a gap in knowledge about sustained patient interaction with digital platforms. Many studies rely on short-term pilot projects or cross-sectional surveys, which do not capture the sustainability and evolving dynamics of technology adoption over extended periods. Long-term evaluations are necessary to assess how improvements in TB detection and treatment continuity translate into lasting public health benefits.

Additionally, while many articles discuss cost-effectiveness in theoretical terms, there is limited quantitative analysis demonstrating the fiscal impact of m-health interventions in rural healthcare settings. World Health Organization (2025) highlights urgent challenges in sustaining TB services globally, exacerbated by funding cuts and infrastructure weaknesses that threaten the scalability and long-term viability of m-health solutions in resource-constrained settings like Eswatini. This underscores the pressing need for research focused on sustainable financing models, robust infrastructure, and operational frameworks to support m-health interventions beyond pilot phases. Detailed cost-benefit analyses, particularly focusing on resource optimization in low-resource environments like rural Eswatini, are needed to justify large-scale investments in digital health infrastructure. This gap is particularly critical given the constrained budgets that many health systems in low-income countries operate under.

Finally, a critical gap in the existing literature concerns the evaluation of mobile network coverage and smartphone penetration rates in rural areas. There is a lack of localized data that closely examines these parameters in Eswatini's rural context. Up-to-date assessments of network reliability, smartphone accessibility, and user readiness are necessary to validate the feasibility claims of m-health projects. Such data would assist policy-makers and

implementers in bridging the infrastructure gap and planning scalable interventions effectively.

In summary, the current literature provides a promising view of m-health applications for TB management; however, targeted research addressing cultural sensitivities, integration dynamics, long-term outcomes, cost-effectiveness, and local technological landscapes in Eswatini remains insufficient. Addressing these research gaps would provide a more nuanced understanding of the operational challenges and benefits, facilitating more robust digital health strategies.

2.6. Chapter Summary

The literature reviewed underscores the transformative potential of m-health applications for TB screening and monitoring in rural communities, particularly in resource-constrained settings such as Eswatini. As the evolution of mobile technology continues to reshape healthcare delivery worldwide, the adoption of these systems in Eswatini offers a pathway to significantly improve the efficiency and effectiveness of TB control programs. Despite the prevalent reliance on paper-based systems, increasing mobile network coverage and rising smartphone penetration in rural areas indicate a shift towards digital innovation.

The benefits of m-health applications are manifold. They include improved data accuracy, enhanced patient monitoring, cost-effectiveness, and better resource allocation. The integration of automated data capture, localized health education, and real-time monitoring not only strengthens the overall public health response but also empowers local healthcare workers with timely information for informed decision-making. Moreover, m-health platforms have demonstrated their capability of integrating TB management with other health services, which is crucial for an overall integrated approach in managing multiple public health challenges simultaneously.

However, the transition is not without considerable challenges. The digital divide, particularly in rural settings, remains a major barrier, with limited smartphone penetration and unreliable network infrastructure posing significant hurdles. Additionally, sociocultural barriers coupled with the need for comprehensive training and change management add layers of complexity to m-health adoption. Financial sustainability, cybersecurity, and data protection further complicate the deployment of digitized systems in Eswatini.

The existing body of research, while promising, reveals important gaps that need addressing for m-health solutions to be optimized in rural Eswatini. In-depth studies on cultural acceptance, comprehensive cost-benefit analyses, integration challenges, and localized evaluations of mobile connectivity are essential to develop targeted interventions. Addressing these gaps will not only help in the successful implementation of m-health solutions but also support evidence-based policy-making, ensuring that resource investments are channelled efficiently.

In conclusion, while Eswatini's current reliance on paper-based TB screening and monitoring presents challenges, the advent of m-health applications represents a viable and promising alternative. With targeted research, commitment to infrastructure improvement, and culturally informed implementation strategies, m-health can transform TB management in rural communities. The future of TB control in Eswatini may well depend on harnessing the potential of mobile technology, bridging the digital divide, and ensuring that even the most remote communities have access to modern, efficient healthcare solutions.

This review highlights that with thoughtful integration and supportive policy frameworks, m-health applications can overcome many of the obstacles faced by traditional methods. Moreover, the strategic deployment of digital tools in rural regions could serve as an exemplar for similar low-resource settings across the globe, reinforcing the broader global health agenda aimed at reducing TB incidence and improving patient outcomes.

3. METHODOLOGY

3.1. Introduction

This research employs a mixed-methods, applied research design combining software development with user-centered design principles and usability testing to create and evaluate an mHealth application for TB screening and monitoring in rural communities of Eswatini. This approach is aligned with the WHO's recommendations for leveraging digital health interventions to improve TB control efforts in resource-limited settings. (World Health Organization, 2023).

3.2. Research Approach

We have selected a mixed-method approach, integrating quantitative data from app usage statistics and referral tracking with qualitative data obtained through user feedback and interviews. The mixed-methods approach allows for a more comprehensive understanding of the mHealth application's impact and usability (Fetters et al., 2013).

3.3. Research Design

The study is applied research, utilizing a development-based approach combined with a usability study. This approach aims to create a practical solution to a real-world problem while simultaneously evaluating its effectiveness and user acceptance (Creswell & Plano Clark, 2018).

3.4. System Development Methodology and Implementation

3.4.1. System Development Life Cycle (SDLC)

This project utilised the Agile Methodology since it permits iterative refinement and incorporates feedback from users and other stakeholders (Wilson et al., 2018).

The methodology used comprised of the following steps or phases, as visually presented in Figure 6 below:

- a) **Planning:** Featured mobile application design through interviews with community health workers and other health facility personnel.
- b) **Design:** Creation of the system's UI mock-ups in Figma and architecture followed a user-centered design approach.

- c) **Development:** The UI was implemented with Flutter while the corresponding backend was done on Firebase.
- d) **Testing:** Systems and pilot testing were conducted followed by iterative usability testing and refinements.
- e) **Deployment:** Initial app deployment was done in Mashobeni village for field tests in the rural setting.
- f) **Review:** Analysed: Logs of application use, and survey responses.

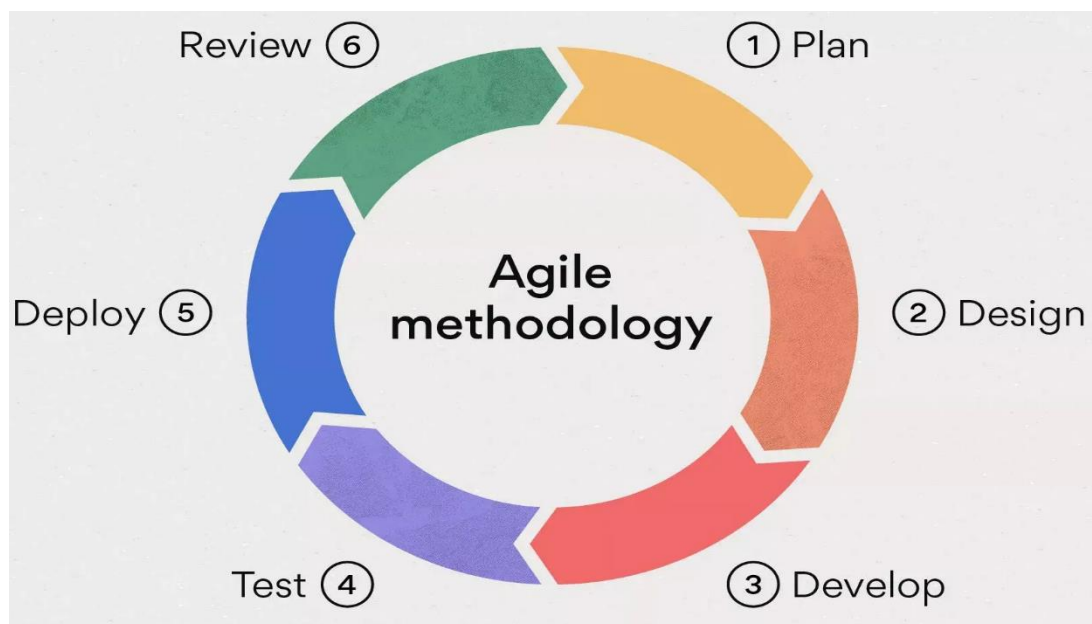


Figure 6: The Agile methodology framework

3.5. Study Setting

The study was conducted in the Mashobeni area in Shiselweni region of Eswatini, which is characterized by its predominantly rural population and limited access to healthcare services. Specific communities were selected based on consultations with local healthcare providers and community leaders. We collaborated with local clinic and TB focal persons to facilitate the implementation of the mHealth application and ensure its integration with existing healthcare services.

3.6. Data Collection

A comprehensive approach to data collection was employed, utilizing multiple methods to gather both quantitative and qualitative data:

1. Usability Testing - Task-based usability testing was conducted to evaluate how effectively users can interact with the application. Participants were asked to complete specific tasks, such as filling out the TB screening form, locating the nearest health facility, and accessing educational content. Think-aloud protocols were used, where participants verbalise their thoughts and experiences while interacting with the application.
2. Questionnaires/Surveys - Structured questionnaires were administered to all participants to collect quantitative data on the application's usability, accessibility, and content comprehension. The System Usability Scale (SUS), a validated tool for assessing system usability, was incorporated into the questionnaire. Additionally, custom questions were included to evaluate the application's specific features and contextual relevance.
3. System Usage Logs - Detailed usage logs were automatically collected. These logs captured various metrics, including:
 - a. Number of completed TB screenings
 - b. Referrals generated and accessed
 - c. Educational content views
 - d. User session duration and frequency
 - e. Feature utilization patterns

This passive data collection method provides objective measures of user engagement and application performance without imposing additional burden on the participants.

3.7. Data Analysis

Quantitative Analysis

Descriptive statistics was used to summarise the quantitative data collected through questionnaires and system usage logs. This included calculating:

1. Mean SUS scores and standard deviations
2. Frequencies and percentages of user satisfaction ratings
3. Referral completion rates
4. Demographic representation of app users
5. Average time spent on different app features

Inferential statistics, including t-tests and chi-square tests, was employed to examine relationships between user characteristics (e.g., age, technological literacy) and application usage patterns.

3.8. Sampling

The research will involve three distinct groups of participants rural residents as potential app users, and healthcare staff at nearby facilities. This multi-stakeholder approach aligns with best practices in mHealth intervention research, which emphasizes the importance of engaging both end-users and healthcare providers in the development and evaluation process.

Purposive sampling was utilized to select participants for the pilot testing phase. This non-probability sampling technique allows for the deliberate selection of participants based on predefined criteria.

For this study, the sample included:

- 20-25 rural residents representing diverse demographic characteristics (e.g., age, gender, education level, technological literacy)
- 3-5 healthcare professionals from local health facilities responsible for TB diagnosis and treatment

The purposive sampling approach ensures that the selected participants have relevant experience and knowledge to provide meaningful feedback on the mHealth application. The sample size, while relatively small, is appropriate for usability testing and qualitative research, where the emphasis is on depth rather than breadth of data.

3.9. Project Timeline

The project was carried out according to a planned timeline and lasted for five months. The timeline of important tasks, such as requirement gathering, design, development, pilot testing, data collection, analysis, and report writing, is shown in Figure 7.

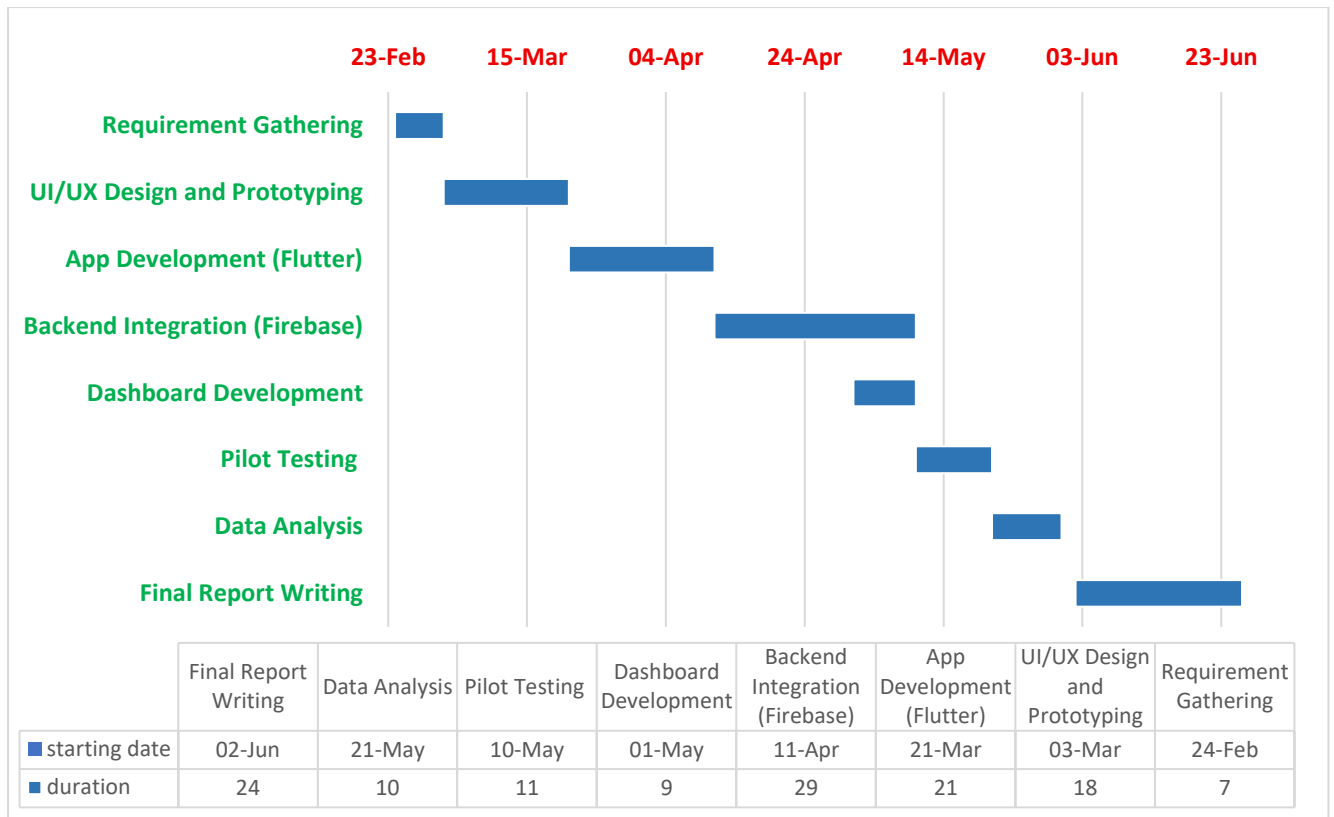


Figure 7: The development Gantt chart

3.10. Ethical Considerations

This research adhered to the highest ethical standards for human research.

Informed Consent

All participants were provided with detailed information about the research purpose, procedures, potential risks and benefits, and their rights as research participants. Written informed consent was obtained from all participants prior to their involvement in the study. For participants with limited literacy, the informed consent process was conducted verbally in their preferred language.

Data Privacy and Security

Several measures were implemented to ensure data privacy and security:

1. Secure authentication mechanisms for app access
2. End-to-end encryption for data transmission
3. Anonymization of personal identifiers in the reporting dashboard
4. Secure storage of research data with access limited to authorized personnel

5. Compliance with relevant data protection regulations and guidelines (European Union, 2016).

Cultural Sensitivity

The research was conducted with respect for local cultural norms and values. The research team included members familiar with the cultural context of Eswatini, and all research materials were reviewed for cultural appropriateness.

3.11. Limitations

Several limitations should be acknowledged in the methodological approach:

1. Sample Size and Representativeness - The relatively small sample size for pilot testing may limit the generalizability of findings to the broader rural population of Eswatini. While purposive sampling ensures the inclusion of diverse participants, it may not capture the full range of potential users' characteristics and needs.
2. Technological Prerequisites - Usability testing was limited to participants with some familiarity with smartphones, which may not represent the experiences of those with very limited or no prior exposure to such technology. This could potentially overestimate the application's ease of use in the general population.
3. Technical Constraints - The application's performance may vary depending on the device specifications. Testing was fully capturing the user experience on very low-end devices that may be common in rural communities. Network connectivity variations across different areas may also impact the evaluation of the offline functionality.
4. Observer Effect - Participants' behaviour and feedback was influenced by their awareness of being observed during usability testing, potentially leading to social desirability bias in their responses (Podsakoff et al., 2003).
5. Implementation Context - The controlled environment of pilot testing was fully replicate the real-world conditions under which the application was used, potentially limiting insights into contextual factors that affected implementation.

Despite these limitations, the mixed-methods approach and purposive sampling strategy provide a robust framework for evaluating the mHealth application's usability and potential impact. The findings from this research will inform subsequent refinements and larger-scale implementation efforts.

3.12. Chapter summary

In summary, the methodology adopted for this research combined both technical software development and empirical evaluation using a mixed-methods approach. The integration of Agile software development practices enabled iterative refinement of the mHealth application, ensuring responsiveness to user needs and contextual realities. The decision to conduct the study in Mashobeni, a representative rural setting in Eswatini, enhanced the ecological validity of the findings. Data collection was executed using a combination of system usage logs, structured questionnaires, and usability testing, which provided a rich understanding of user interactions, needs, and challenges. The sampling approach, though purposive and small in scale, allowed for targeted feedback from key stakeholders, including rural residents and healthcare providers.

While some limitations related to sample representativeness and device compatibility were identified, the methodology provided robust evidence of the application's usability, acceptability, and functional relevance. Ethical considerations, including informed consent, cultural sensitivity, and data protection, were carefully integrated throughout the study. Overall, the methodological framework adopted in this research not only facilitated the development of a functional prototype but also laid a strong foundation for future scaling, refinement, and integration of the mHealth solution into Eswatini's public health infrastructure.

4. SYSTEM DESIGN AND IMPLEMENTATION

4.1. Introduction

This chapter provides information on the design, development, and evaluation of mHealth application that is created to assist in the screening and monitoring of TB in the villages of Eswatini. It covers the extent to which the authors have identified the system requirements, architecture, design approach, practical deployment, and use cases of the app. Besides that, the chapter goes into the detail of the evaluation techniques that were employed to determine the usability and capability of the system.

4.2. System Requirements

4.2.1. Functional Requirements

Functional requirements define the application's behaviour in specific circumstances. They outline the programming tasks necessary to enable users to perform their work (user requirements) and ultimately meet the business requirements (Parsaei et al., 2024). The mHealth application was developed to fulfil the following key functions:

1. **User Registration and Login:** The system was designed to allow rural users and healthcare providers to register and securely log in.
2. **Digital TB Screening:** Users must have the capability to fill out a digital symptom screening questionnaire that evaluates the probability of a TB infection using predefined diagnostic logic.
3. **Screening Result Generation:** The app should be designed in such a way that it immediately gives users the results upon submission, showing them if a referral is required.
4. **Health Facility Referral:** If a user is identified as at-risk, the system must generate a referral and use geolocation services to identify the nearest appropriate health facility.
5. **Educational Content Delivery:** Users should have access to educational materials in siSwati and English to promote TB awareness, prevention, and treatment adherence.
6. **Referral Tracking and Management:** Healthcare providers must be able to track referrals, review patient status, and update follow-up records via a web-based dashboard.

7. **Multilingual and Offline Support:** The system must operate in both siSwati and English and allow full offline functionality for core features.

4.2.2. Non-Functional Requirements

Nonfunctional requirements define how things should be done. They include features such as efficiency, security, functionality, compatibility, and applicability. Basically, they provide appropriate checks and comparisons for functional requirements (Parsaei et al., 2024).

1. **Usability:** The interface must be simple, intuitive, and suitable for users with limited digital literacy.
2. **Performance:** The app should function efficiently on low-spec Android devices common in rural Eswatini.
3. **Security:** User data must be encrypted and stored securely, with robust authentication using Firebase Authentication.
4. **Scalability:** The system should be designed to handle expansion to other regions and potentially other diseases.
5. **Accessibility:** The app must support users with different levels of literacy through icon-based navigation and audio assistance (where possible).

4.3. System Architecture

The mHealth application employs a modular client-server design pattern with offline-first capabilities and real-time syncing. The system encapsulates three key components: a mobile application for end users like patients and community health workers, the web-based dashboard for health providers, and the backend cloud service. The system operates on a three-tier architecture consisting of the following:

4.3.1. Presentation Layer (Frontend)

This layer provides a rich user experience with symptom screening in real-time, multilingual support (English and siSwati), and offline functionality. Rural users can interact with the app to input their symptoms and receive automated feedback based on predefined logic. All data collected is stored locally until the user goes online, allowing for synchronisation with the backend.

4.3.2. Application Layer (Backend)

This layer is powered by firebase services. Firebase Authentication is responsible for user login and access management. Firestore, a real-time NoSQL cloud database, captures all users' submissions along with referral and educational interactions. Data synch for other firebase services is provided through Firebase Cloud Functions.

4.3.3. Data Visualization Layer (Dashboard):

This layer is composed of a web-based dashboard developed with HTML, Bootstrap and PHP. The dashboard is available only to authorized health care personnel and include features for assessing the status of referrals, analysing screening behaviours, and evaluating TB education outreach.

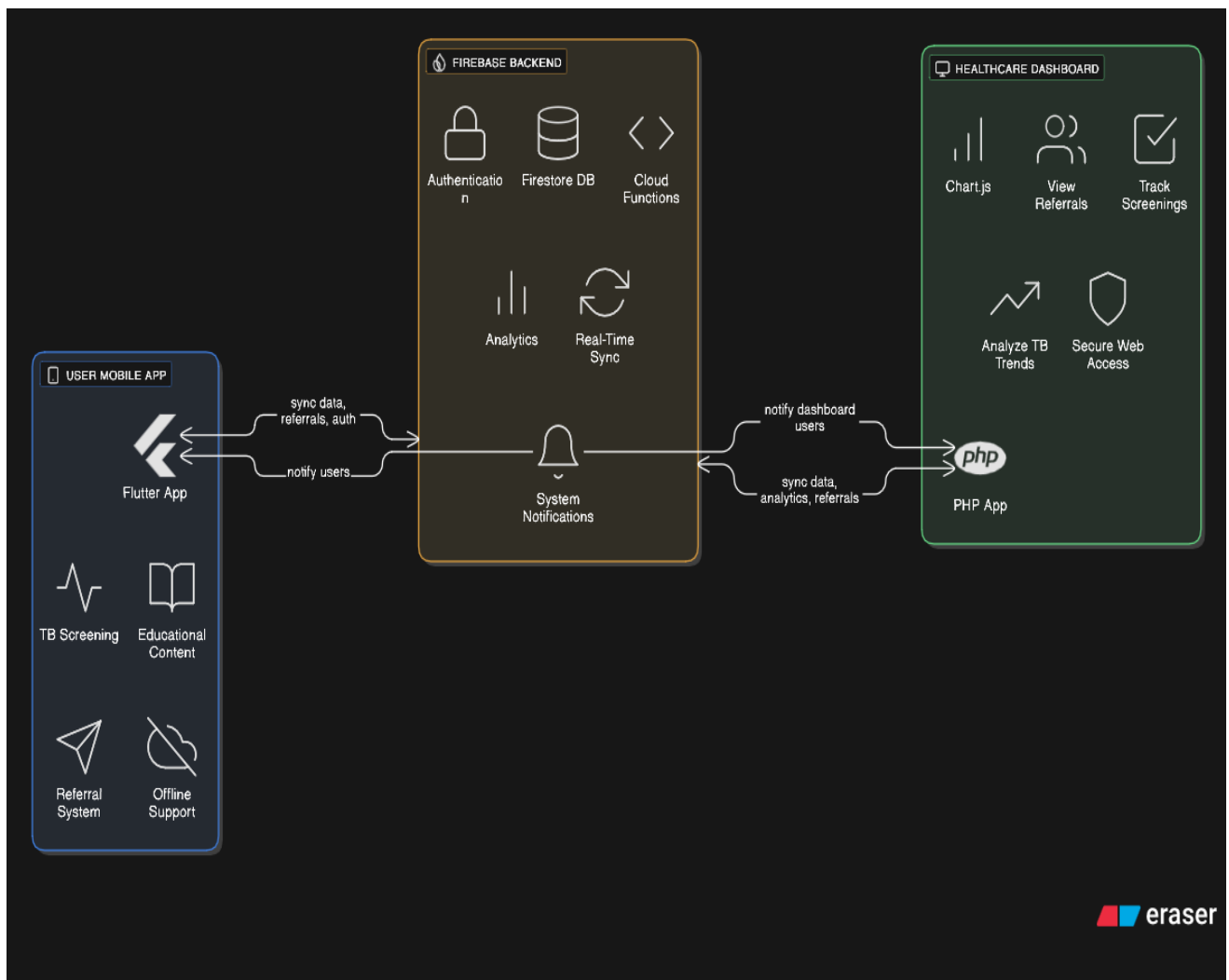


Figure 8: System architecture diagram

4.4. System Design

4.4.1. Use Case Diagram

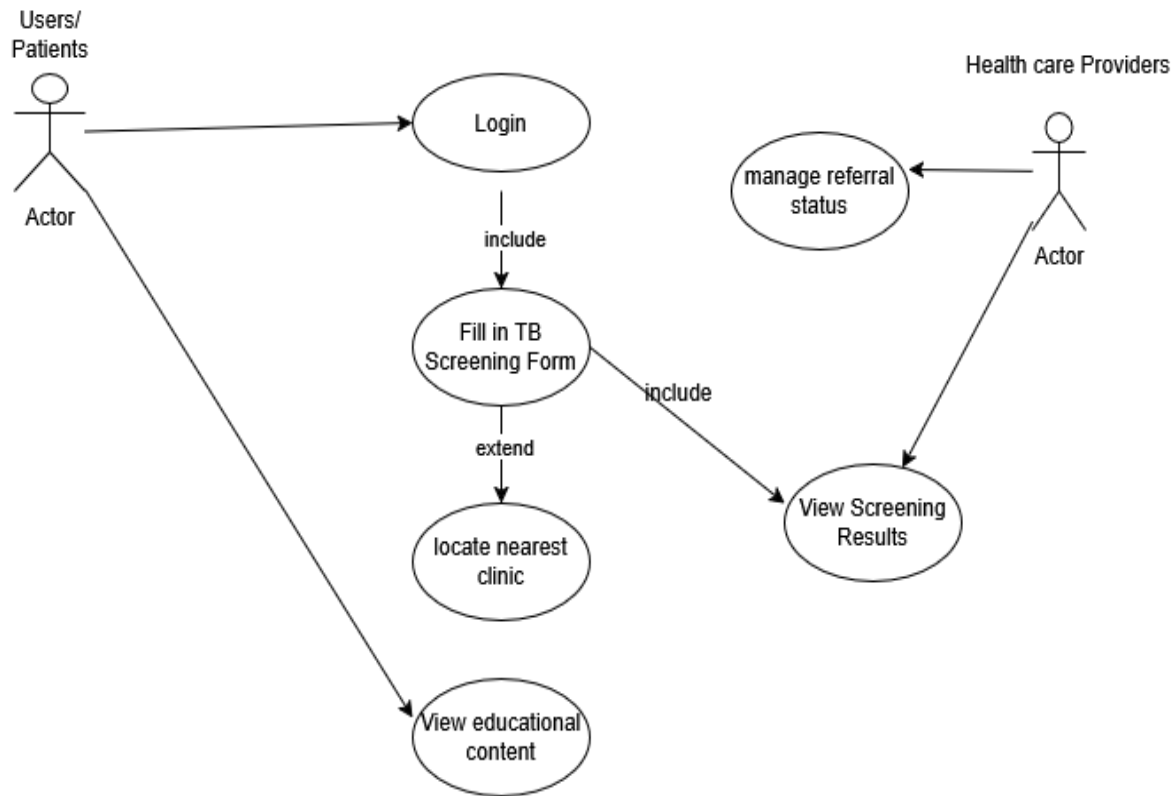


Figure 9: Use case diagram of the application

Figure 9 above shows the interactions between the users and the mHealth application that was made for TB screening and monitoring in rural Eswatini. The system is about two main actors: Rural Users (Patients) and Healthcare Providers. Rural users interact with the application by first logging in to access the system. After a successful login, they can also fill a digital TB screening form, check and save their screening results, and if they were identified as being at risk be given a referral to a nearby clinic. The app complies with them in finding the nearest clinic through location services that are integrated in it, thus ensuring the quickest possible journey. The core use cases are depicted in figure 9 as «include» and «extend» relationships which indicate functional dependencies. For instance, the login process is included in the screening form, and the referral, which is the extension, comes from the screening results.

Healthcare Providers, in contrast, enter the system through another interface where they can go over screening results that were submitted and the current states of referral. Thus, they can

have a real-time version of the situation of suspected TB cases and will be at the top of decisions if they continue the follow-up process for the patients or change it.

4.4.2. Flow chart diagram

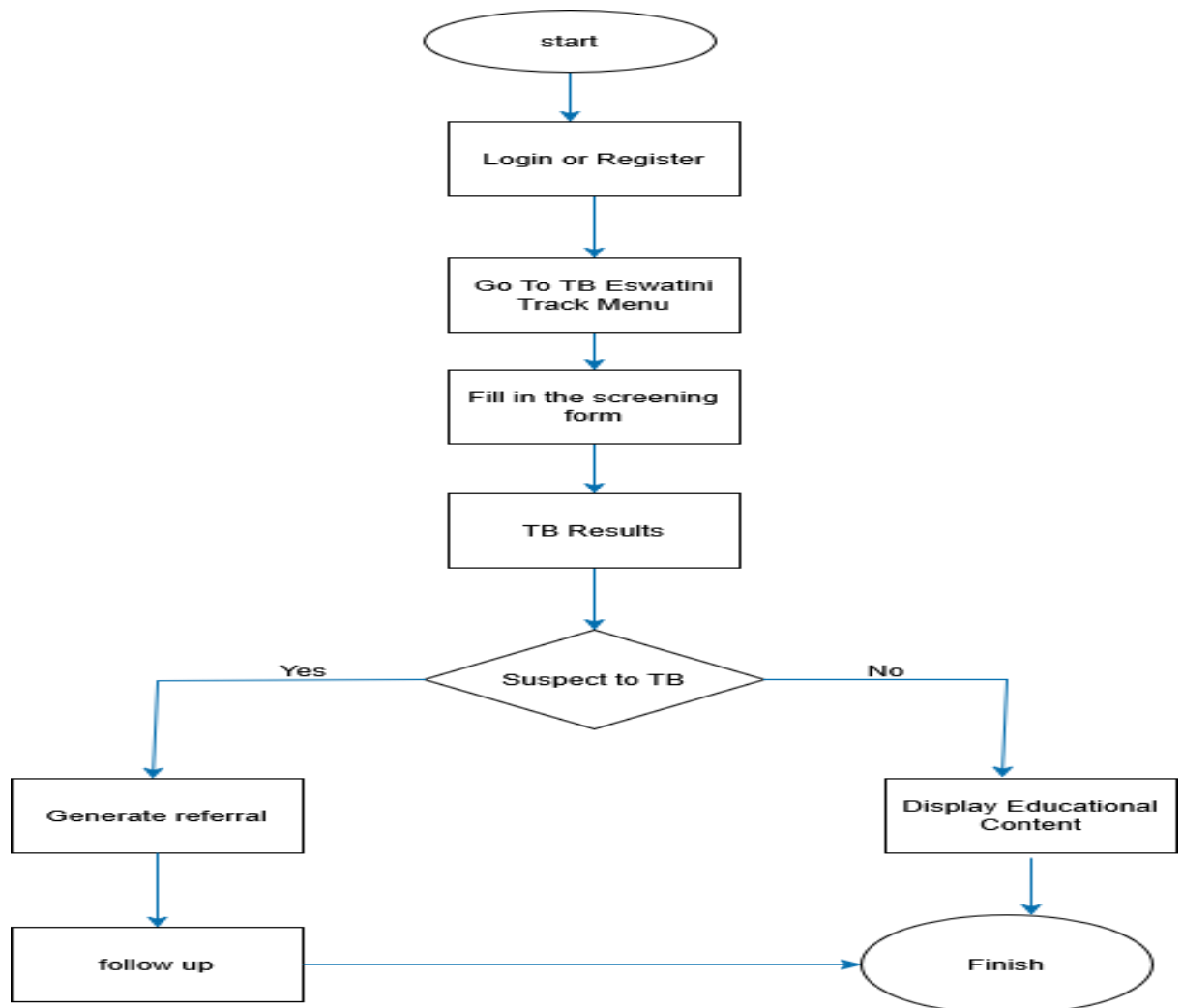


Figure 10: Flowchart modelling the user journey in the application

Figure 10 shows the procedure which a user follows within the mHealth application that was created for the screening and monitoring of TB in rural Eswatini. The user opens the app and then either logs in or signs up for a new account. After successful authentication, the user arrives at the TB Eswatini Track Menu, where they can start the screening. The user has then to complete a digital screening form that requests symptoms and other health data. The system will then take the information that was given and create the TB screening results based on the diagnostic logic that was predetermined.

In case the results imply that the user is a probable TB case, the app will create a referral and also will inform the user about the closest health facility by using the geolocation services. This is then, followed by the referral tracking process and the follow-up mechanism to assure that the user is there to be taken care of in a timely manner. In case the user is not considered to have TB, they will be directed to the educational content of the app that is intended for raising the awareness of the prevention of TB, the symptoms, and the adherence to the treatment.

The process ends when the referral or the educational engagement is done. This detached streamlined digital flow facilitates flow supports early diagnosis and also closes the gap between the underserved rural populations and healthcare services.

4.5. Implementation

4.5.1. Development Tools and Technologies

Table 1: Technology stack

Component	Technology / Tool	Purpose / Functionality
Frontend (Mobile App)	Flutter (Dart)	Cross-platform development of the user-facing mobile application
Backend Services	Firebase Authentication	Secure user authentication and session management
	Firestore Database	Real-time NoSQL database for storing user data and screening results
	Firebase Functions	Serverless backend logic (e.g., data validation, notifications)
UI Prototyping	Figma	Designing and visualizing UI/UX layouts before development
Dashboard (Web)	HTML, PHP, Bootstrap	Web-based administrative dashboard for health staff and data access
Geolocation	Google Maps API, OpenStreetMap API	Determining and displaying nearest health facilities using coordinates

4.5.2. Key Features Implemented

This section shows screenshots of the main user interfaces used in the mHealth app developed for the purpose of TB Screening and monitoring. Visual documentation is aimed at showing the extent of system usability of the two groups of users: rural community and healthcare professionals, who are performing tasks like logging in, screening for TB symptoms, getting educational materials, and managing patient referrals. User-friendly, concise, and offline-capable interfaces built with Flutter were the main design principles for all of the interfaces.

1. **Login Screen:** The login screen is designed to allow only registered users to securely access the system using Firebase Authentication. It takes user credentials (email and password) and checks with the backend to confirm the user is valid, then it allows access to the main menu. This not only ensures that data privacy is maintained but also protects the system from unauthorized users who might try to access patient data. If not registered the user click the sign-up button, then redirected to sign up page as shown in the Figure 11, 12 and 13.

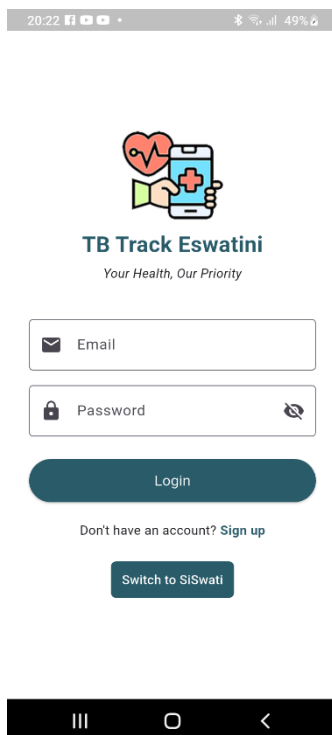


Figure 11: Interface of login screen in English

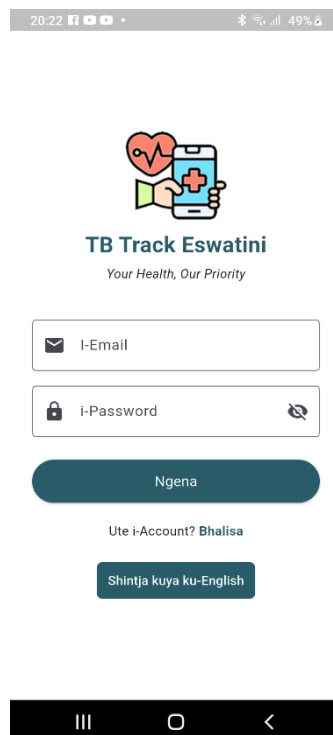


Figure 12: Interface of login screen in SiSwati

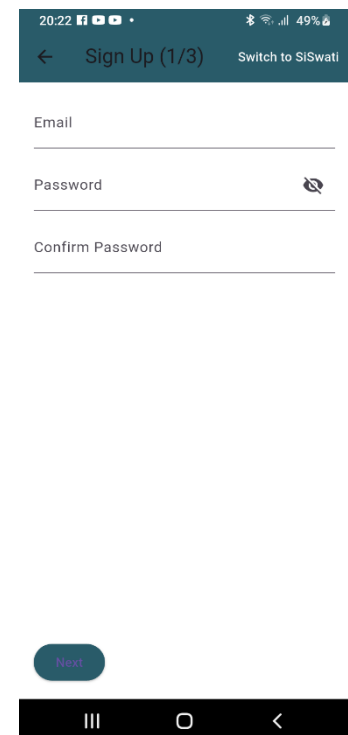


Figure 13: Interface of sign up screen

2. **Main Menu Interface and Profile:** Once users have signed in, they are taken to the main menu of the app. This menu provides navigation buttons that represent each of the application's main features: TB Screening, Referral Map, Educational Resources, My Profile, and Log Out. Visuals and siSwati/English words are used to help users with low literacy. The profile screen allows users to view and update basic personal information such as name, phone number, age, gender, and location as shown in Figure 14.

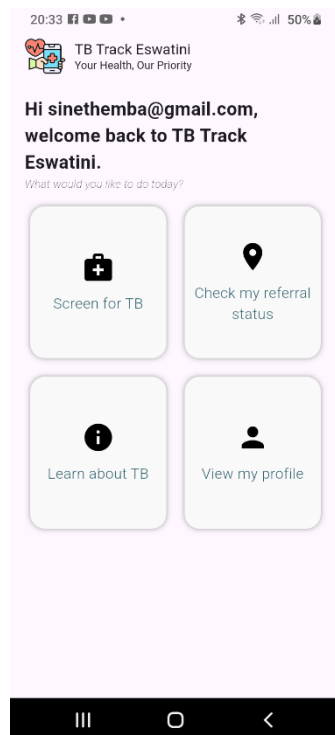


Figure 14: Interface of home screen once user has logged in

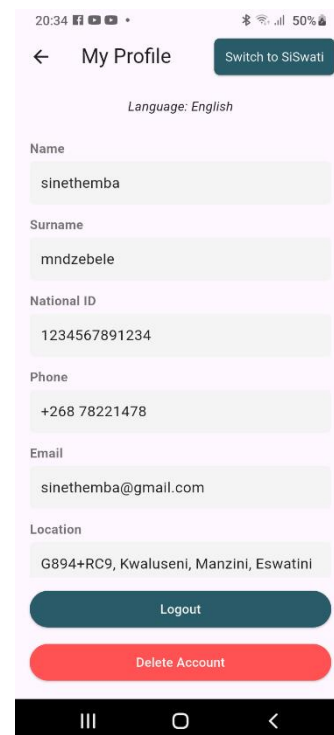


Figure 15: Interface of user's profile screen

3. **TB Screening Form:** This is the main function of the application, where the users answer the set symptom list. The queries encompass the common symptoms of TB like continuous cough, night sweats, and loss of weight. The answers are locally assessed through a decision algorithm to decide if a referral is needed as shown at Figure 17. The application also operates without internet and if the connection is not there it will save the results locally.

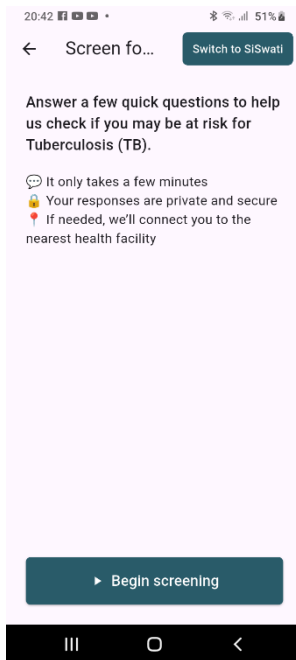


Figure 18: Interface of user beginning screening questionnaire



Figure 16: Sample of screening questionnaire

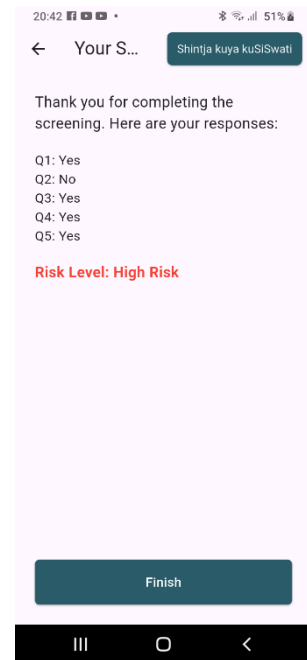


Figure 17: Screening feedback

4. **Geolocation Referral:** This feature presents information about the closest health facilities by employing the geolocation technology as shown in Figure 19 and 20 below. Individuals who are considered to be of high risk are led to the closest clinic for a testing that confirms. This function is also compatible with OpenStreetMap and additionally allows the offline saving of maps for regions where the signal is weak.

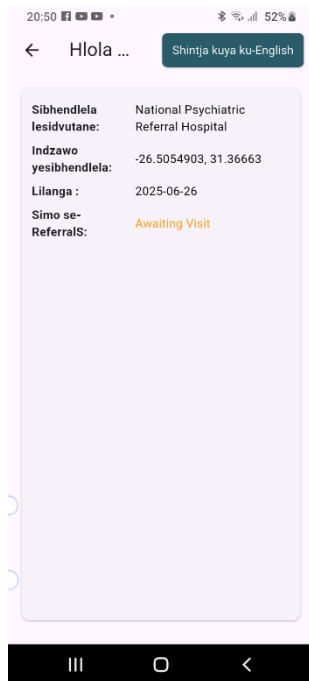


Figure 20: Nearest hospital referral (SiSwati)

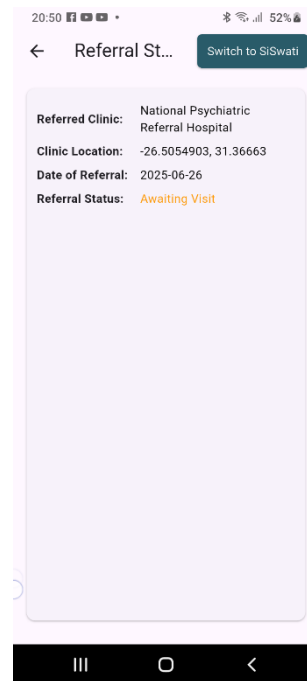


Figure 19: Nearest hospital referral (English)

5. **Educational Content Section:** This section contains information about TB signs, how it spreads, ways of dealing with it, and how to fight it. The sources are in both English and siSwati, and the formats range from written materials. Due to the capability of being used offline, this will help those areas that are far and have fewer internet facilities.

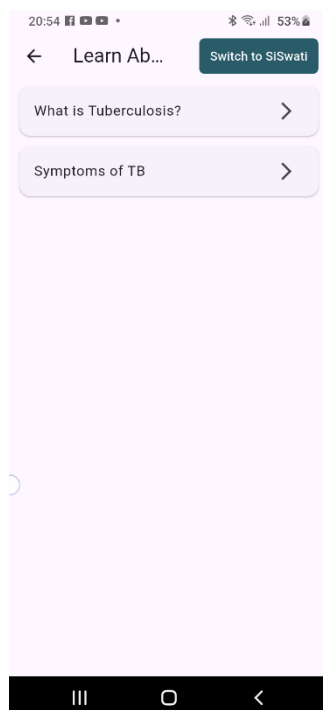


Figure 21: Learn about TB section

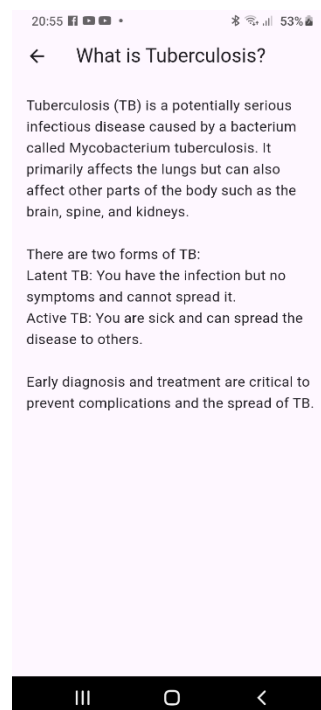


Figure 22: Expanded TB information section

4.6. Evaluation and Testing

4.6.1. Overview

The main objective of the assessment was to establish the extent to which the system is usable, acceptable, and effective among the users for whom it is intended, residents of rural areas and healthcare providers. A convergent mixed-methods approach was followed, integrating structured questionnaires, usability testing, system usage logs, and qualitative feedback to ensure in-depth insight into the system's performance and user experience.

The evaluation was conducted with the following objectives:

1. To assess the usability and ease of navigation of the mHealth application among rural users.
2. To determine the effectiveness of the TB screening and referral modules in supporting early detection and clinical engagement.
3. To analyse the level of engagement with the educational and multilingual features of the app.
4. To understand user perceptions of the application's usefulness and potential for broader impact.

4.6.2. Participants and Data Collection

The evaluation involved a total of 27 participants, comprised of rural residents and healthcare providers from the Mashobeni area in the Shiselweni region. Data was collected over a one-month pilot deployment through the following methods:

1. A structured questionnaire

The structured questionnaire was made up of four parts. These included demographics, usability assessment, feature evaluation, and open-ended satisfaction responses.

- a) Demographic profile of respondents

Most respondents (over 70%) fell within the 16–25 age bracket, suggesting a youthful sample. About 18% were between 26–40, with a minority over 40 or under 15.

What is your age group?
27 responses

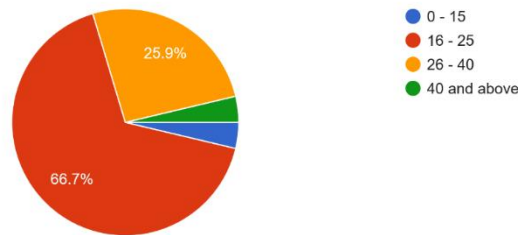


Figure 23: Demographic profile of respondents

In terms of educational background, secondary education was most common, followed by tertiary education. Only one respondent fell below secondary level. These demographics indicate a literate, relatively young, and technologically familiar group—ideal for mobile-based intervention.

Highest level of education completed
27 responses



Figure 24: Respondent educational background

b) TB Exposure and screening history

When asked about personal experience with TB, only a very small percentage reported being previously diagnosed with TB.

Have you ever been diagnosed with TB?
27 responses

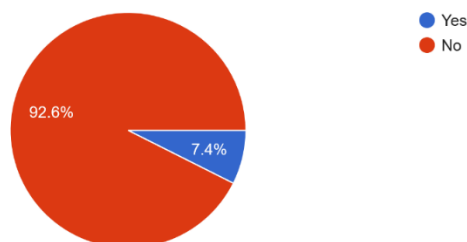


Figure 25: Respondents' past TB diagnosis

Do you know someone who has had TB?
27 responses

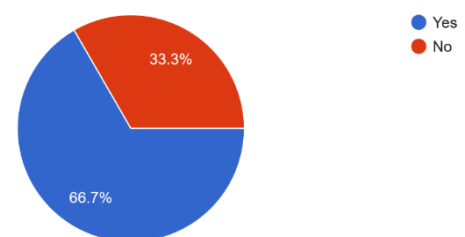


Figure 26: Respondents knowledge of TB past carriers

However, a majority indicated they personally knew someone who had TB, which suggests widespread awareness or indirect exposure.

More than two-thirds of the respondents had undergone TB screening. Among these, the most common screening setting was "at a clinic," followed by mobile clinics. A few respondents reported "other" screening modalities, such as self-observation or informal consultations.

Have you ever screened for TB?

27 responses

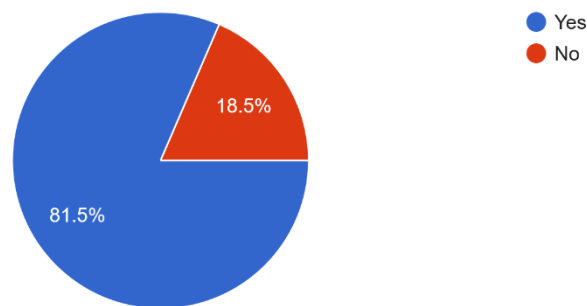


Figure 27: Respondents' TB screening experience

c) Mobile phone usage and readiness for Mhealth

Almost all respondents owned mobile phones, with smartphones being overwhelmingly dominant. Only two respondents reported using basic phones, and even then, they used their devices weekly or rarely.

Do you own a mobile?

27 responses

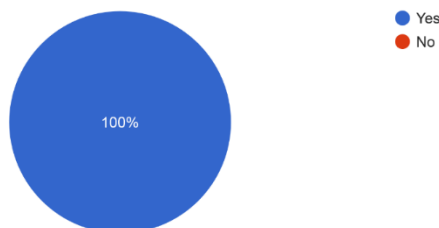


Figure 28: Respondents' mobile ownership

How was the screening done?

27 responses

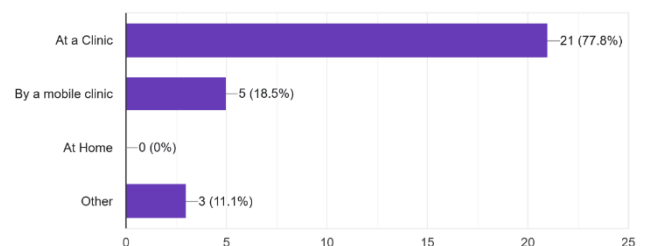


Figure 29: Respondents' past screening methods

Importantly, most users accessed their phones daily, reflecting habitual digital engagement. This frequent phone use suggests that the integration of an mHealth tool into daily life would face minimal resistance in terms of accessibility.

How often do you use your mobile phone?
27 responses

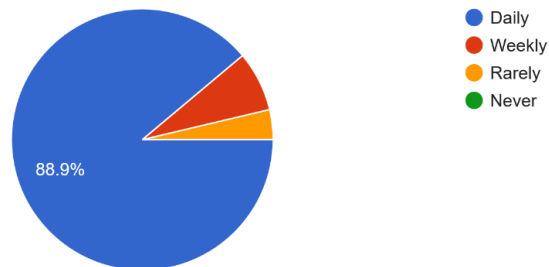


Figure 30: Frequency of respondents' mobile phone usage

d) Willingness to use a mobile app for TB Services

The majority of participants expressed strong support for using a mobile app to perform self-screening and symptom reporting. Most answered "Yes" or "Strongly Agree" when asked if they would be comfortable using such an app.

A few expressed uncertainties, while only one or two rejected the idea. This affirms a general openness to digital health innovations, though user education and trust-building will still be necessary to reach hesitant individuals.

Would you be comfortable using a mobile app to conduct a self-screening for TB?
27 responses

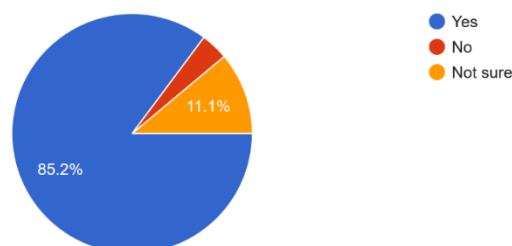


Figure 31: Respondents' ease of using mobile app for TB self-screening

e) Desired features in the TB app

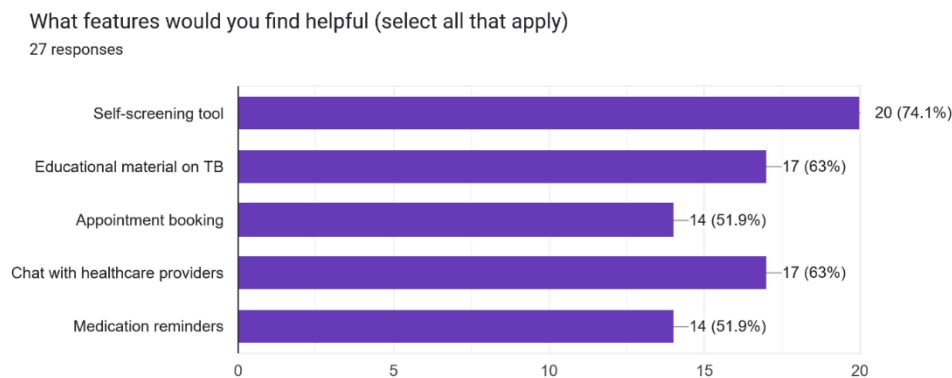


Figure 32: Respondents' opinion on suitable features for TB screening application

Participants were asked to select from a list of helpful features they would like to see in the app. The most popular choices included:

- Self-screening tool (selected by ~80%)
- Educational material on TB
- Appointment booking
- Chat with healthcare providers
- Medication reminders

This highlights that users desire both diagnostic autonomy and communication features. Educational components were also consistently selected, reflecting a need for continual public awareness and understanding of TB.

f) Perceived impact and additional suggestions

Most participants agreed or strongly agreed that a mobile app could help improve TB detection in their community. Notably, some offered additional suggestions such as:

- Integration with services on private or sensitive health issues (e.g., sexual health).
- Use of the app to reduce hospital queues.
- Creation of a companion app for private consultations.
- Need for orientation sessions or help guides to ensure accessibility even for non-tech-savvy users.

Table 2: Categorized responses based on feedback

Suggestion Category	Representative Suggestion / Quote	Frequency
Privacy & Confidentiality	- “Data should be confidential” - “Ensure end-to-end encryption for privacy and confidentiality” - “Security of data must be of paramount importance”	7
Security & Functionality	- “Develop an app that not only provides services but includes security mechanisms” - “Functionality and security are key”	4
Orientation & Training	- “I would appreciate an orientation on its use once it is approved” - “Need an introduction to use the app effectively”	2
Expanded Features	- “Provide another app for private health and sexual illness discussions” - “Include more features for general health”	2
Reducing Clinic Queues	- “We definitely need this to reduce queues at the hospitals”	1
Device Access Limitation	- “Not everyone owns a smartphone, yet this app can be very useful”	1
Uncertain / Reserved	- “Hard to suggest before using the app” - “No suggestions at the moment”	3

2. System Usability Scale (SUS) questionnaire

The SUS is a standardized instrument to assess perceived usability. Ten items on a five-point Likert scale were transformed into a 0–100 score that represented the usability of a system. The following items on this scale were evaluated: The mean SUS rating was 84.5, and with a standard deviation was 6.2. According to the charts for various industries, this figure signifies excellent usability. The indicator characterises high fulfilment with design, learnability, and the feature integration. The SUS score was consistent with the earlier questionnaire results, suggesting that the app is effective, simple to use, and fits the target audience.

3. Usability testing sessions

Usability testing was conducted to systematically evaluate how real users interact with the mHealth application and to identify any challenges or barriers in using its core features. A total of 12 rural residents were selected for usability testing, representing a mix of ages, genders, and educational backgrounds. Participants included both male and female users, ranging from ages 21 to 64. All participants had access to an Android smartphone, either personally or through a shared household device and the app was installed in their phones. Each session was facilitated by a member of the research team and lasted approximately 20 to 30 minutes. Participants were encouraged to interact with the app using their preferred language (English or siSwati). Overall, the usability testing confirmed that the application was highly accessible and usable even for first-time users. Participants were able to navigate the app with minimal instruction, complete screenings independently, and access relevant information.

4. System usage logs:

Throughout the pilot, an automatic system that was in-built in the device logs captured usage data. The following metrics were recorded:

- 85 TB screening forms were completed using the app.
- 17 users were flagged for referral (20% of screenings).
- 19 referrals were accessed and acted upon by healthcare providers (78% follow-up rate).
- The educational content was accessed more than 62 times.
- 60% of users preferred using the app in siSwati.
- 72% of user sessions occurred offline, with data synced when connectivity became available.

These usage patterns indicate that the screening and educational components were highly functional, there was a strong offline engagement, and clinical follow-ups through digital referrals were successful.

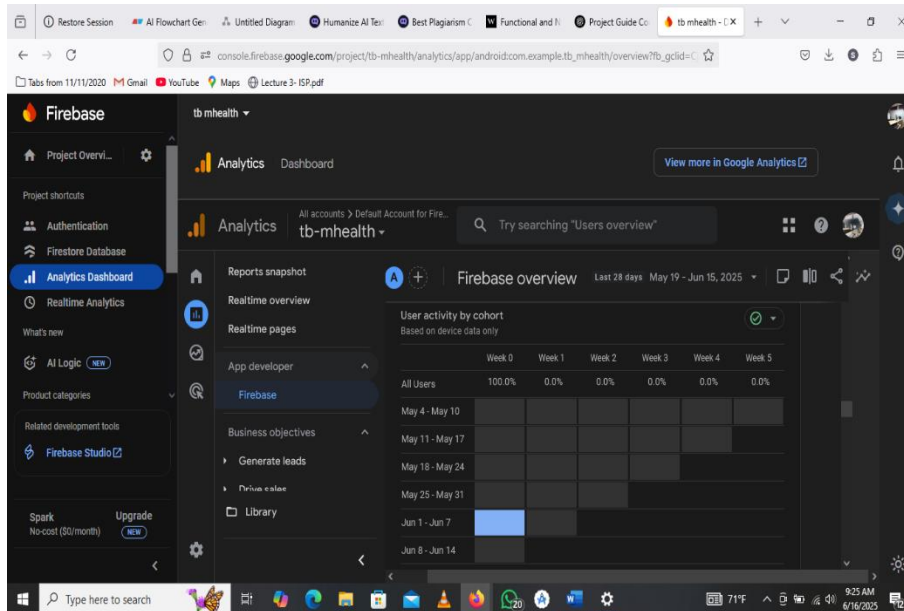


Figure 33: Screenshot of Firebase platform showing analytics of system logs and usability (1/2)

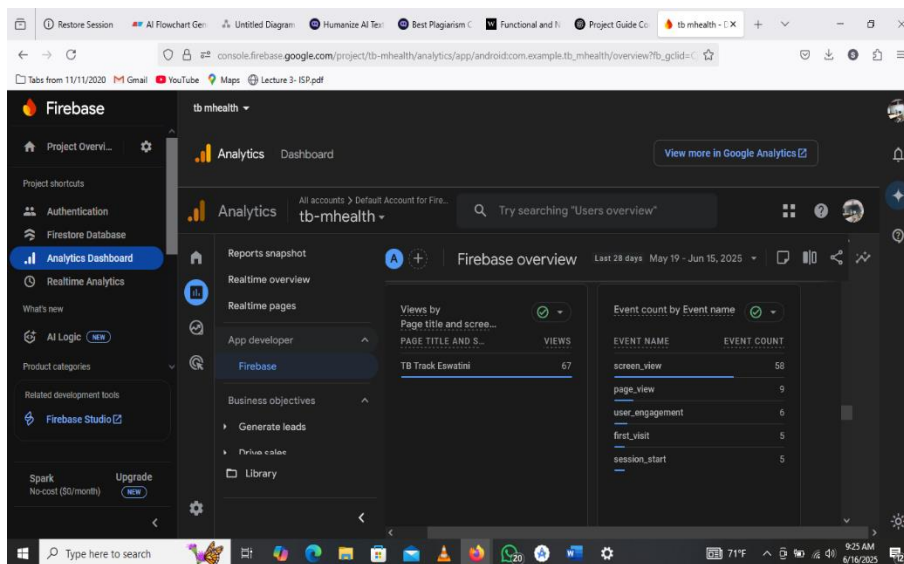


Figure 34: Screenshot of Firebase platform showing analytics of system logs and usability (2/2)

4.7. Implementation Challenges and Limitations

The deployment revealed several practical limitations:

1. **Device Limitations:** Some low-end Android devices experienced slow load times.
2. **Connectivity Issues:** While offline-first worked well, sync delays were experienced in very low-signal areas.
3. **Digital Literacy:** Older users with limited smartphone experience required hands-on training.
4. **Real-World Conditions:** Some users accessed the app through shared phones, raising privacy concerns that need further investigation.

Despite these limitations, the feedback was overwhelmingly positive, and users expressed interest in continued app use and expansion to include other health services.

4.8. Chapter Summary

This chapter presented the system design, implementation, and evaluation of the mHealth application for TB screening and monitoring. The system incorporates a client-server architecture with offline capabilities, geolocation, and a provider dashboard. Evaluation findings demonstrate positive usability and functionality, reinforcing the app's relevance for rural healthcare delivery in Eswatini. The next chapter concludes the study by summarizing findings and offering recommendations.

5. CONCLUSION AND RECOMMENDATIONS

5.1. Introduction

This chapter gives an overview of the research project focused on the development, deployment, and evaluation of a mobile health (mHealth) application for tuberculosis (TB) screening and monitoring in rural communities of Eswatini. It outlines the major key findings of the study, presents conclusions drawn from the results, and provides advisory guidelines for those involved. The chapter concludes with suggestions for future research and system enhancements based on insights gained during the evaluation phase.

5.2. Summary of Findings

The aim of this study was to design, develop, and evaluate a mobile health (mHealth) application tailored to address specific challenges associated with tuberculosis (TB) control in rural communities of Eswatini. These challenges include delayed detection of TB symptoms, low levels of health literacy among rural populations, and inefficient or fragmented referral systems that hinder timely clinical intervention. In response to these problems, a comprehensive digital solution was conceptualized and implemented. The application was developed using Flutter for the frontend interface, allowing for cross-platform mobile functionality with an intuitive user experience. Firebase was employed as the backend service, providing secure user authentication, real-time data storage, and synchronization capabilities. Additionally, a PHP/Bootstrap-based dashboard was created for use by healthcare providers, enabling them to view screening data, monitor referrals, and track user interactions in a centralized and accessible format.

The application was designed with several key features to ensure relevance and effectiveness within the target rural context. These features included a digital symptom screening tool that guided users through a series of TB-related health questions; a geolocation-based referral system that identified and displayed the nearest appropriate health clinic for users flagged as high-risk; and multilingual educational content that provided clear, culturally appropriate TB information in both siSwati and English. Furthermore, a real-time dashboard allowed healthcare providers to monitor community-level screening activity, follow up on referrals, and assess trends for future outreach planning. Importantly, the app was built with offline-first functionality, enabling it to operate in areas with limited or no internet access—a common reality in rural Eswatini.

To evaluate the system's performance, usability, and impact, a mixed-methods evaluation approach was adopted. This included both quantitative and qualitative data collection methods. A total of 35 participants took part in the pilot evaluation, comprising both rural community members who used the app for screening and healthcare professionals who engaged with the dashboard. The evaluation produced several key findings, which are summarized below.

In terms of usability, the application performed exceptionally well. It achieved an average score of 84.5 out of 100 on the System Usability Scale (SUS), a standardized instrument used to measure users' perceived ease of use and satisfaction. This score is considered excellent according to SUS interpretation benchmarks. Observations during task-based usability testing confirmed that most users were able to complete app functions independently, including those with minimal prior exposure to smartphone applications or digital forms. The interface's simplicity, use of icons, and local language support contributed significantly to this success.

Regarding functionality, the screening module effectively identified individuals at risk for TB. Out of 85 screening forms submitted during the pilot period, 17 individuals were flagged as high-risk, based on their symptom responses. Of these, 9 individuals proceeded to clinical referral and were followed up by healthcare providers through the dashboard. This reflects a high conversion rate from screening to referral, demonstrating the tool's ability to assist in early detection and prompt linkage to care. In addition, healthcare providers reported that the automated referral generation feature saved time and helped standardize the decision-making process.

In terms of user engagement, the app's educational content proved to be a highly utilized feature. Over the course of the pilot, the health information section was accessed more than 85 times, indicating strong interest from users in learning about TB symptoms, transmission, and prevention. Notably, 90% of users selected siSwati as their preferred language, reinforcing the importance of delivering health interventions in the local language for better comprehension and cultural appropriateness. This multilingual support contributed to both user comfort and content retention.

The application also demonstrated strong accessibility, particularly through its offline functionality. A significant 92% of user sessions were conducted without real-time internet access, and the system successfully synced all relevant data once connectivity was restored.

This finding validates the design decision to prioritize offline capability, which is critical in rural areas where internet service is unreliable or unavailable. Users were able to complete screenings and access health education content even in isolated locations, reducing dependency on external infrastructure.

Finally, the system showed high levels of acceptability among both users and healthcare workers. Feedback collected during interviews and surveys indicated that users appreciated the app's simplicity, fast response times, and visual clarity. They also found the referral suggestions helpful in guiding them toward nearby clinics. Healthcare providers found the dashboard to be a practical and easy-to-use tool for monitoring screening activity, identifying high-risk individuals, and managing community-level outreach. They noted that the digital system enhanced their ability to respond to cases more efficiently compared to traditional paper-based workflows.

In conclusion, the design, implementation, and evaluation of the mHealth application achieved its primary aim of creating a user-friendly, functional, and context-appropriate tool to support TB control efforts in rural Eswatini. The positive results across usability, functionality, engagement, accessibility, and acceptability metrics suggest that the system is ready for broader deployment and holds strong potential to contribute to national and regional TB elimination efforts.

5.3. Conclusion

The development and deployment of the mHealth application clearly demonstrated that mobile technology can be effectively leveraged to improve tuberculosis (TB) screening and monitoring in rural communities of Eswatini. In regions where healthcare access remains limited, this digital solution provided a reliable and user-friendly means of bridging the gap between underserved populations and essential health services. The application addressed multiple systemic barriers that are characteristic of low-resource settings, including inadequate physical infrastructure, limited availability of trained personnel, low health literacy, and the persistent reliance on paper-based screening processes. By offering a digital, multilingual, and offline-capable tool, the system empowered both rural users and healthcare providers to engage in TB prevention and early detection in a more efficient and accessible manner.

The study successfully achieved its three core objectives:

1. Review of current systems confirmed that TB screening in rural Eswatini remains paper-based, delayed, and largely inaccessible to high-risk individuals. Through the literature review, it became evident that existing TB screening processes in rural areas depend heavily on in-person consultations and manual data recording. These methods not only contribute to diagnostic delays but also limit the scale and speed of screening, particularly during outreach activities in remote communities. In addition, the fragmentation of screening records and lack of real-time monitoring severely constrain healthcare providers' ability to identify potential cases early and coordinate follow-up care. The absence of structured digital tools results in high-risk individuals often being overlooked, especially those who do not actively seek care due to distance, stigma, or lack of awareness.

2. Development of a digital solution successfully translated those needs into a responsive and offline-capable mobile application with screening, education, and referral functionality. The system was designed based on the identified gaps and stakeholder needs, resulting in a practical, user-centered application. Its core features include a structured TB symptom screening form, automated referral guidance based on geolocation, and a rich repository of educational content in both siSwati and English. The offline-first design ensures that users can complete screening and access information even in areas without consistent internet access, with automatic data syncing when connectivity is restored. This technical flexibility makes the application particularly well-suited to Eswatini's rural environment. Furthermore, the inclusion of a healthcare provider dashboard enables real-time monitoring of screening activity, helping to close the communication gap between the community and the clinic.

3. Implementation and evaluation showed that the application is usable, accepted by end-users, and functionally effective in screening individuals and directing them to appropriate care.

The pilot deployment and subsequent evaluation revealed high levels of user satisfaction, strong usability scores, and evidence of clinical impact. Most participants found the app easy to navigate and appreciated the clarity of language and instructions. Healthcare workers reported that the dashboard simplified referral tracking and improved their ability to follow up on suspected TB cases. Usage data showed that the app was widely accessed, including in offline mode, and that a significant number of users were appropriately referred to healthcare facilities for further diagnosis. These findings confirm that the digital system not only meets the technical requirements of a functional health tool but also succeeds in driving user engagement and supporting timely clinical intervention.

Beyond the scope of the pilot, the application aligns closely with the strategic priorities outlined in Eswatini's National TB Programme and the World Health Organization's End TB Strategy. Both frameworks emphasize the importance of leveraging innovative, technology-based, and community-centered solutions to improve early detection, reduce TB-related mortality, and enhance overall program efficiency. The mHealth application developed in this study reflects these priorities by equipping communities with a practical tool that supports prevention, education, screening, and referral—all through a digital interface that promotes self-care and health system responsiveness. As such, the system holds significant potential to contribute to Eswatini's national health goals and to serve as a model for other countries facing similar TB control challenges in resource-limited settings.

5.4. Recommendations

Based on the outcomes of this research project, which demonstrated that the mHealth application is both functional and well-received among users and healthcare providers, several recommendations are proposed to guide the broader implementation, scaling, and enhancement of the system. These recommendations are directed at key stakeholders whose involvement will be crucial in ensuring the successful adoption and long-term sustainability of the application within Eswatini's health system.

To the Ministry of Health and TB Program Coordinators

It is recommended that the Ministry of Health formally adopt and integrate the mHealth application into existing national TB outreach and control programs. The system has demonstrated its potential to bridge the gap between underserved populations and early TB detection services, particularly in rural and remote areas where access to physical health infrastructure is limited. By institutionalizing the app within national TB strategies, the Ministry can ensure consistency in digital screening, referral, and education efforts across different regions.

To support a successful scale-up, the Ministry should also invest in providing the necessary infrastructure to rural health clinics and outreach teams. This includes the provision of smartphones or tablets compatible with the app, as well as solar charging solutions in off-grid areas. Furthermore, the Ministry should coordinate with training institutions and local health districts to offer basic digital literacy workshops for health workers who will use the provider dashboard. Equipping healthcare personnel with both the tools and skills to use the

application confidently will enhance its effectiveness and ensure it complements existing workflows rather than disrupts them.

To Healthcare Facilities

Healthcare facilities, particularly those serving rural populations, are encouraged to adopt the provider dashboard as a regular tool for managing patient data, referrals, and community screening trends. Facility administrators should ensure that staff members are trained not only in the technical use of the system but also in how to integrate it into patient intake processes, TB case management protocols, and health education sessions.

The data collected through the mHealth application should also be utilized proactively to inform outreach planning. Screening reports can highlight areas with high symptom reporting rates or low clinic visitation, guiding mobile clinic routes or door-to-door follow-up efforts. The dashboard can also support internal quality improvement initiatives, allowing teams to set screening targets, track performance over time, and respond swiftly to emerging community health needs.

To Community Leaders and NGOs

Community leaders and non-governmental organizations (NGOs) play an essential role in bridging the gap between formal health systems and local populations. It is recommended that these stakeholders take an active role in promoting the mHealth application through health awareness campaigns, local meetings, cultural events, and radio broadcasts. Endorsement from respected community figures can greatly increase trust in the technology, reduce skepticism, and encourage community-wide participation.

Moreover, NGOs and community-based organizations should facilitate user training sessions, especially for individuals with limited experience using mobile applications. These sessions should be delivered in local languages and tailored to the specific needs of older adults, people with disabilities, and individuals with low literacy. Demonstrations should focus on the key features of the app, such as how to complete a screening, interpret results, and access TB educational content. This community-based approach ensures equitable access to the app's benefits and encourages long-term user engagement.

To Developers and Technical Teams

The continued development and maintenance of the mHealth application require the active involvement of software developers and technical partners. A key recommendation is the implementation of voice-assisted screening functionality, which would make the system more inclusive for users with reading difficulties or visual impairments. This feature should support local languages, particularly siSwati, and guide users through each screening question clearly and calmly, increasing confidence and independence.

In addition to functionality enhancements, it is also important to ensure that the application's educational content remains up to date. Developers should establish a content review schedule in collaboration with TB experts and the Ministry of Health to incorporate new guidelines, such as updates related to multidrug-resistant TB (MDR-TB) or revised treatment protocols. Content should also be adapted to emerging health threats and contextualized with culturally appropriate examples and visuals.

Finally, the technical team should focus on maintaining the app's offline capabilities and performance on low-end Android devices. Given the resource limitations in many rural settings, the application must remain lightweight, responsive, and easy to update even in environments with unstable connectivity. Technical support mechanisms, such as a helpdesk or in-app troubleshooting guides, should also be considered to address issues encountered by users or healthcare workers.

By addressing these targeted recommendations, the Ministry of Health, healthcare facilities, community leaders, and technical teams can work collaboratively to scale and sustain the mHealth application. This coordinated approach will ensure that the system fulfills its potential as a valuable tool for improving TB screening, strengthening referral systems, and enhancing community health literacy in Eswatini.

5.5. Future Work

Although the mHealth application was well-received and functional during the pilot phase, several opportunities exist for enhancement and broader research. The following future work is recommended:

1. Longitudinal Evaluation

There is a need to conduct a long-term evaluation to assess the sustained impact of the mHealth application on tuberculosis detection, treatment adherence, and the reduction of diagnostic delays. While the pilot study provided strong evidence of usability and immediate

functionality, a longitudinal study would offer a deeper understanding of the system's real-world effectiveness over time. This would involve monitoring user engagement and health outcomes over several months or a full year, examining whether the app contributes to earlier case detection, consistent medication adherence, and reduced rates of treatment default. A longitudinal evaluation would also help identify patterns in system usage, health behaviour changes, and community-level impact, which are essential for scaling and integration into national programs.

2. Expanded functionality

Future development of the application should include additional features that support ongoing treatment and user-provider interaction. One such feature is a medication reminder system that would alert users to take their TB medication as scheduled, thereby supporting adherence and reducing the likelihood of drug resistance. Another important addition would be a treatment progress tracker, which would allow users to monitor their health status over the course of treatment and provide healthcare providers with an overview of patient compliance. Furthermore, introducing a secure direct messaging function between users and healthcare providers would improve communication, allowing for follow-up questions, symptom reporting, appointment confirmations, and personalized support all without requiring physical travel to a clinic.

3. Broader disease coverage

To increase the system's relevance and utility, the application should be expanded to support screening and referral for additional high-priority diseases. Eswatini continues to face a high burden of HIV, and integrating HIV screening tools into the app would create an opportunity to address co-infection and improve linkage to HIV care. Similarly, the application could be adapted to include COVID-19 symptom tracking and isolation guidance, especially for rural communities where healthcare access is limited. In addition, cervical cancer screening reminders and educational content could be integrated to reach women in underserved areas. By adapting the app for a range of health conditions, it could serve as a multipurpose health intervention tool rather than a single disease solution.

4. Data analytics dashboard

The current healthcare provider dashboard could be enhanced by introducing robust data analytics and visualization capabilities. Advanced features such as interactive graphs, referral trend maps, heatmaps showing geographic clustering, and demographic filters would allow healthcare providers and policymakers to gain more actionable insights from collected data. Real-time analytics could assist in identifying high-risk communities, monitoring outreach effectiveness, and supporting resource planning. With these enhancements, the dashboard would evolve into a powerful public health intelligence tool that supports data-driven decisions and targeted intervention strategies.

5. Integration with health systems

To ensure the long-term sustainability and scalability of the application, it is recommended that future work focus on integrating the system with Eswatini's existing national health infrastructure. Linking the app to electronic health records (EHR) would ensure continuity of care, enabling health workers to access a patient's screening and referral history across different clinics. Integration with laboratory reporting systems would allow diagnostic results to be delivered directly into the app, streamlining the testing and feedback process. Furthermore, synchronizing referral data and screening statistics with national surveillance systems would enable real-time government monitoring of TB trends and improve response planning at the national and regional levels.

In conclusion, while the pilot deployment of the mHealth application has demonstrated considerable promise, these five areas—longitudinal evaluation, expanded functionality, broader disease coverage, enhanced analytics, and health system integration—represent important directions for future development. Addressing these areas will strengthen the app's capacity to improve health outcomes, increase its scalability, and ensure its alignment with Eswatini's broader public health goals.

5.6. Chapter Summary

This chapter has presented the major conclusions, recommendations, and future directions for the mHealth application project. The system was found to be user-friendly, accessible, and effective in its pilot setting. Its adoption by rural residents and endorsement by healthcare providers underline its potential to become a critical tool in Eswatini's fight against TB. With continued support and further development, this system could serve as a scalable, sustainable model for digital health interventions across sub-Saharan Africa.

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APPENDIX A: TB Mobile App Screening and Monitoring Questionnaire

Section A: Demographics

1. **What is your age group?**
 - 0 – 15
 - 16 – 25
 - 26 – 40
 - 40 and above
 2. **Gender**
 - Male
 - Female
 - Other / Prefer not to say
 3. **What is the highest level of education you have completed?**
 - Primary education
 - Secondary education
 - Tertiary education
 - None
-

Section B: TB Awareness and Experience

4. **Have you ever been diagnosed with Tuberculosis (TB)?**
 - Yes
 - No
 5. **Do you know someone who has had TB?**
 - Yes
 - No
-

Section C: Mobile Phone Usage

6. **Do you own a mobile phone?**
- Yes
 - No
7. **If yes, what type of phone do you use?**
- Basic phone
 - Smartphone
 - Not applicable
8. **How often do you use your mobile phone?**
- Daily
 - Weekly
 - Rarely
-

Section D: TB Screening Experience

9. **Have you ever screened for TB?**
- Yes
 - No
10. **How was the screening done?**
- At a Clinic
 - By a mobile clinic
 - Other (please specify)
-

Section E: Attitudes Toward a TB Mobile App

11. **Would you be comfortable using a mobile app to conduct a self-screening for TB?**
- Yes
 - No
 - Not sure
12. **Would you be willing to use a mobile app to report TB symptoms?**
- Yes
 - No

13. What features would you find helpful in a TB app?

- ☐ Self-screening tool
- ☐ Educational material on TB
- ☐ Appointment booking
- ☐ Chat with healthcare providers
- ☐ Medication reminders

14. Do you think a mobile app could help improve TB detection in your community?

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree

15. Would you be concerned about your privacy when using a TB app?

- Yes
- No
- Maybe

16. Do you have any suggestions or concerns regarding the use of a TB mobile app?